New Figure Reconstructions

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Each page places a current book figure beside its new reproducible reconstruction. The figure numbers and complete captions identify the accompanying main book. Continued figures keep their original numbers and identify the continued display. Unnumbered illustrations and the cover are listed separately. Accepted reconstructions have been integrated into the reading editions and are omitted here. This volume contains only figures for which further revisions were requested. Where numbering changed, the previous audit number appears beside the current book number.

These revised reconstructions remain proposals for review. The left column shows their current book illustrations. Use a figure number, together with a panel or continuation when needed, to request changes. The footer identifies the corresponding generation-code directory.

The notes distinguish measured data, controlled numerical experiments, and schematic constructions. Parameters, random seeds, source data, and numerical checks are stored beside each figure.

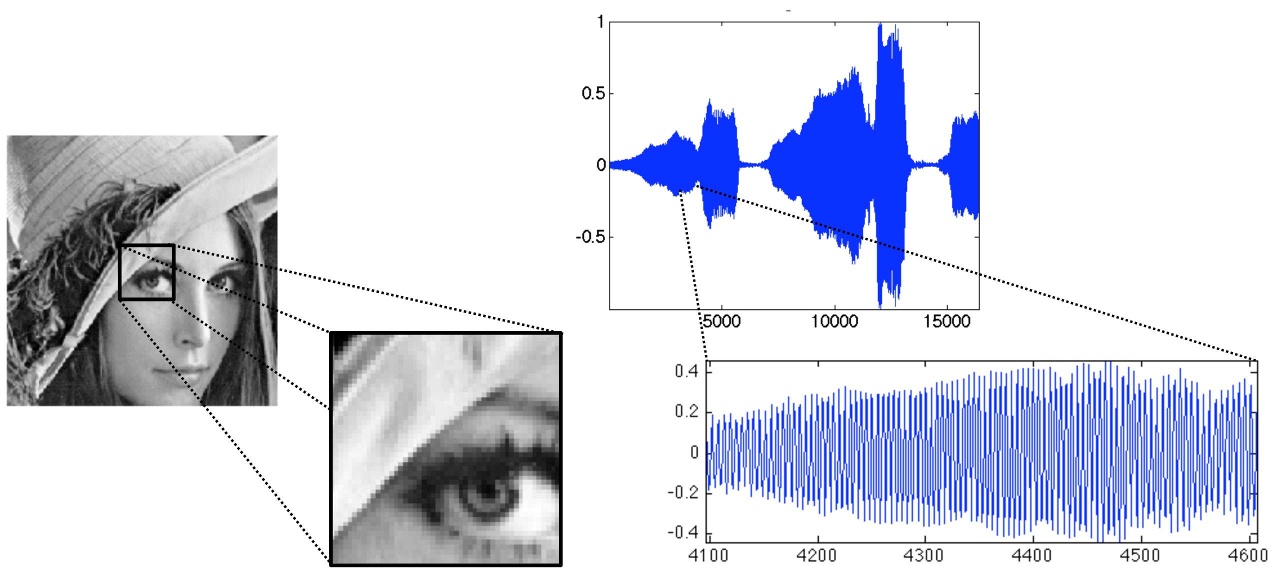
Figure 1.3

Main book, page 3

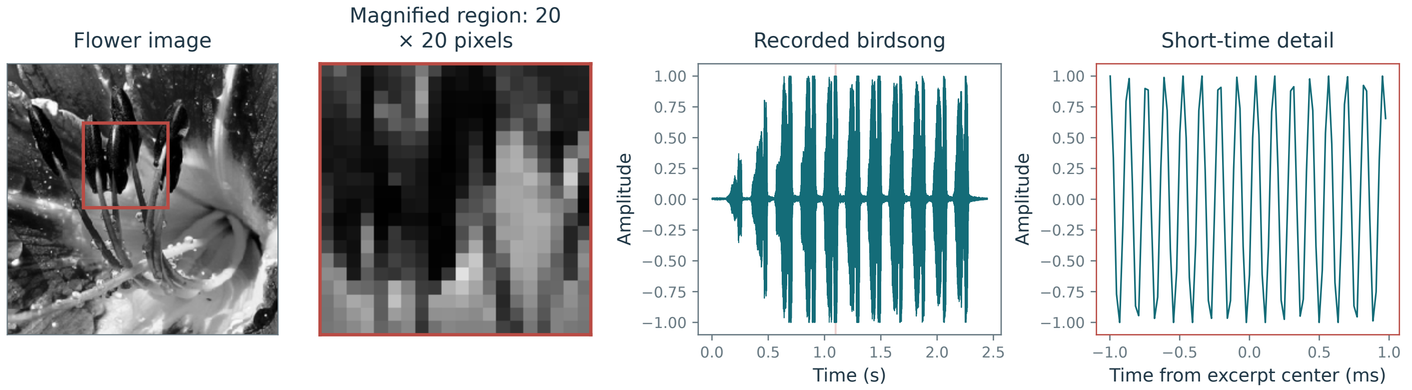
Image and sound discretization.

Shannon Sampling Theory

Current book



Proposed reconstruction



Mathematical and generation notes. The red spatial box selects precisely the flower crop shown as 20 by 20 area-averaged pixels. The sound panels use the supplied bird recording and a 2 ms excerpt around a strong chirp; the red interval marks this exact excerpt. No temporal subsampling is applied.

Code: `figures-code/chapter-shannon-sampling-theory/discretization`

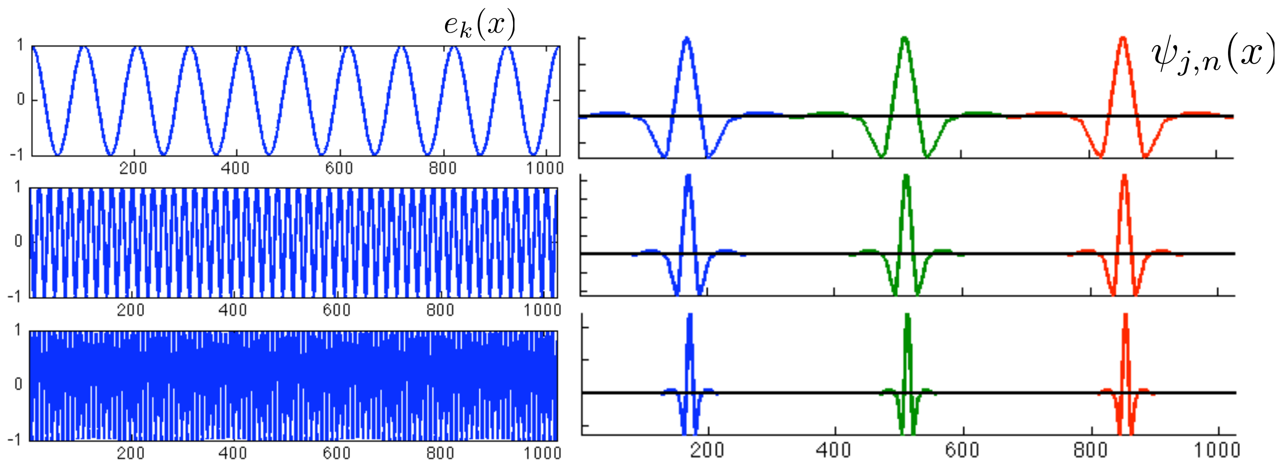
Figure 2.1

Main book, page 10

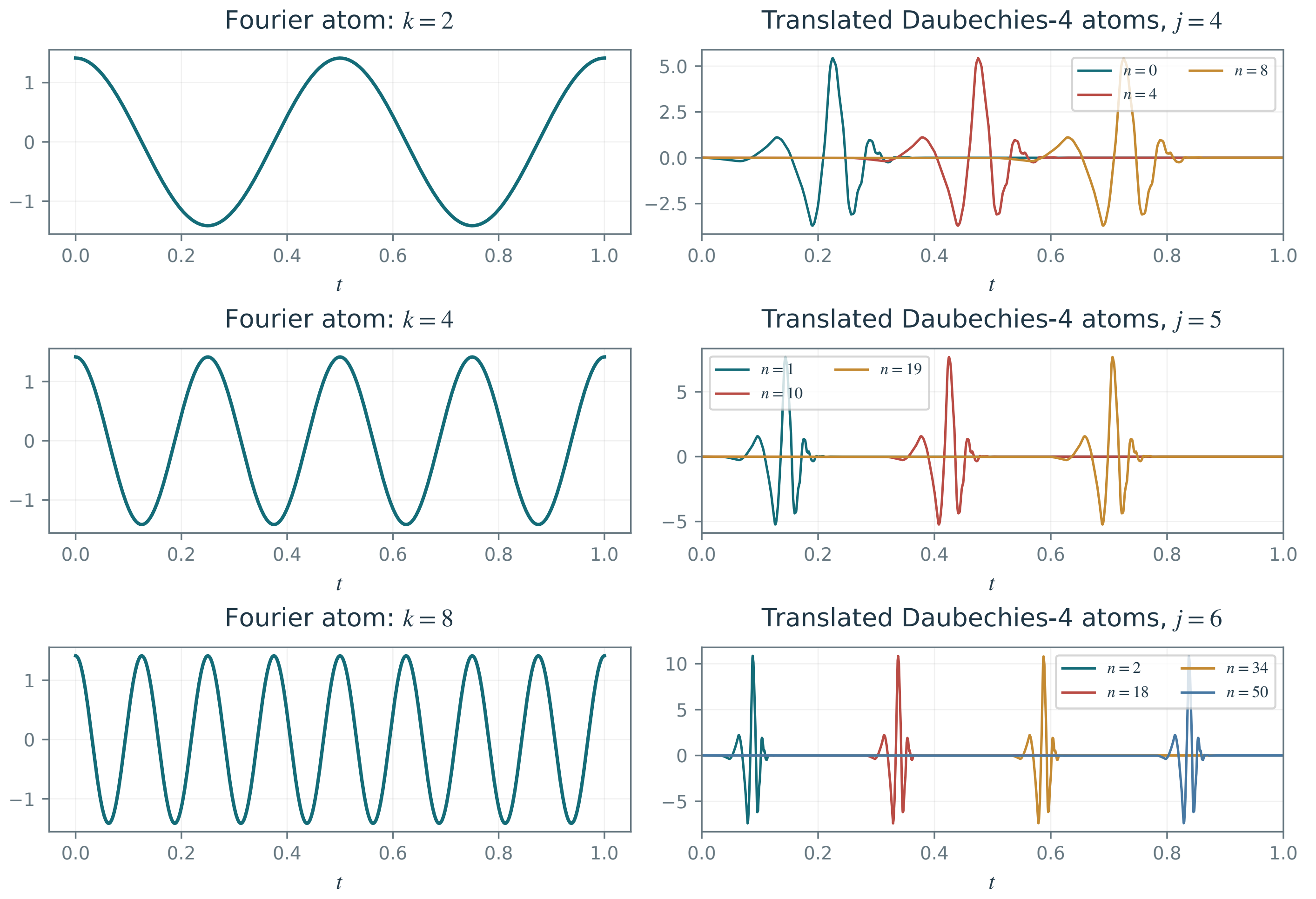
Examples of 1-D Fourier atoms (real parts, left) and wavelets (right).

Fourier and Convolution

Current book



Proposed reconstruction



Mathematical and generation notes. Each right panel overlays several integer translations of the same actual Daubechies-4 wavelet at a fixed scale. The displayed atoms are 2^(j/2) psi(2^j t-n); supports stay inside the common unit interval. Distinct colors identify translations, while successive rows change frequency or scale.

Code: `figures-code/chapter-fourier-and-convolution/fourier-wav`

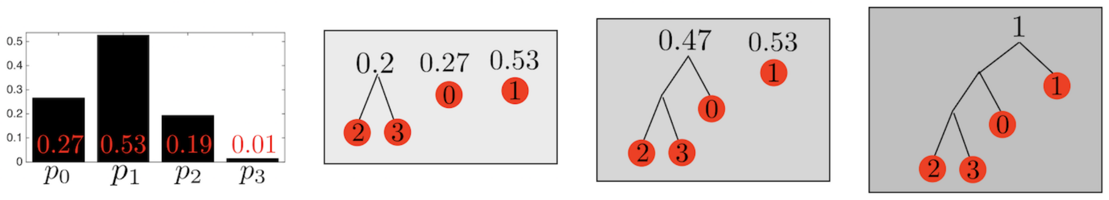
Figure 3.6

Main book, page 37

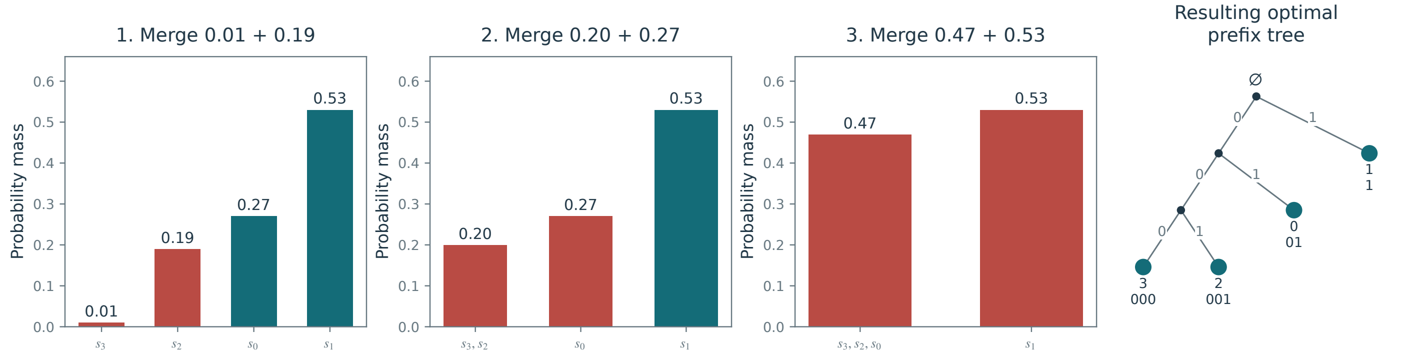
Successive merges in Huffman's coding algorithm.

Shannon Coding Theory

Current book



Proposed reconstruction



Mathematical and generation notes. The merges are computed from probabilities (0.27,0.53,0.19,0.01), rather than inferred from the old drawing. Red bars identify the two least likely current nodes. The resulting tree gives code lengths (2,1,3,3), expected length 1.67 bits, and the correct entropy bound.

Code: `figures-code/chapter-shannon-coding-theory/huff-code`

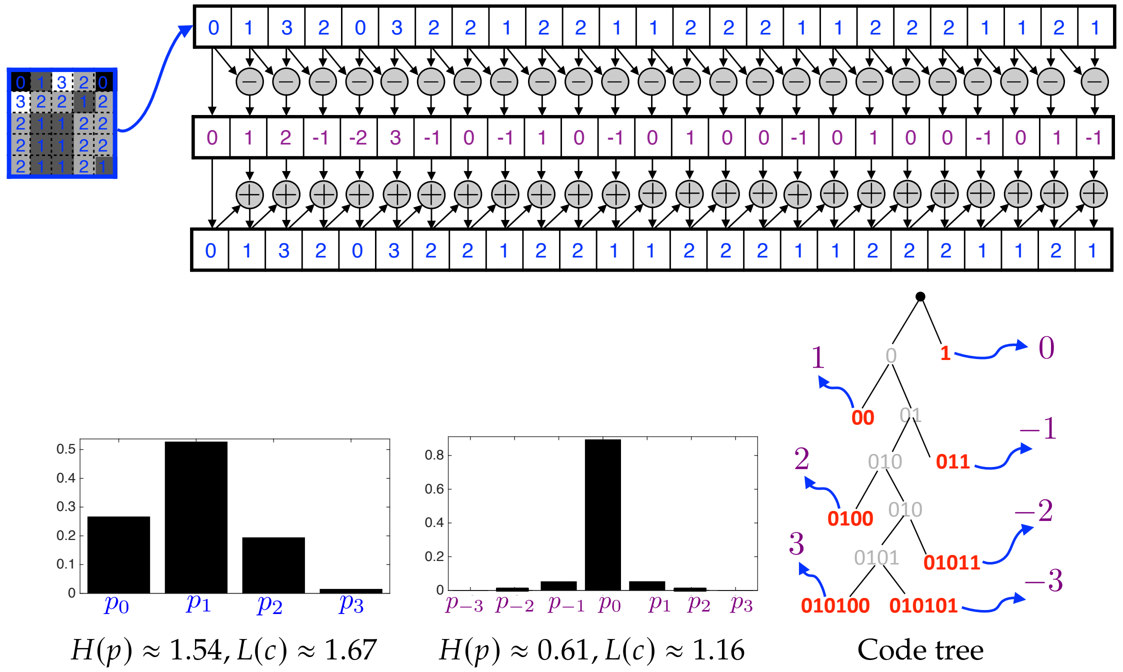
Figure 3.7

Main book, page 39

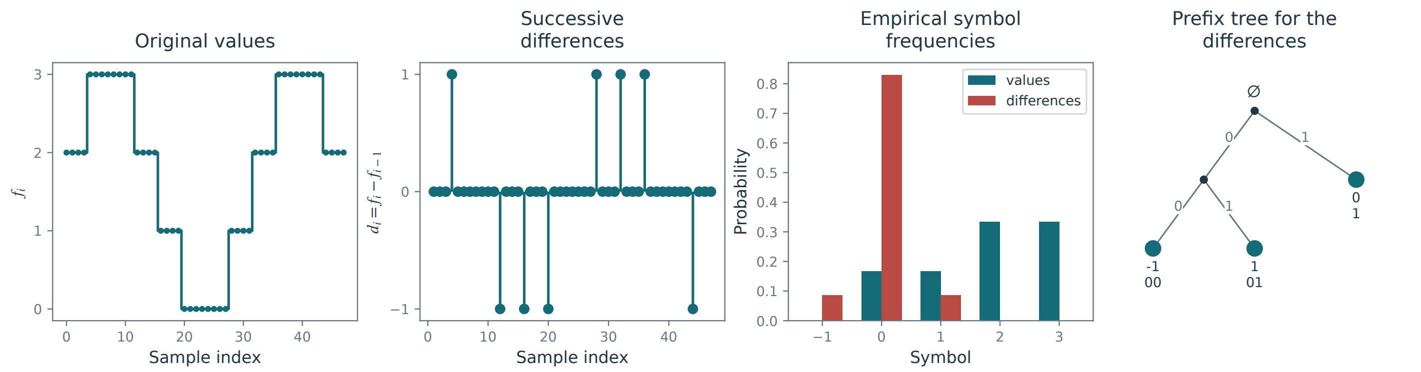
Top: transformation by successive differences. Bottom: comparison of histograms of pixel values and differences, and a code tree for these differences.

Shannon Coding Theory

Current book



Proposed reconstruction



Mathematical and generation notes. A fixed discrete signal is differenced, and both histograms and the Huffman tree are recomputed from its actual counts. Reconstruction retains the initial value and cumulatively sums the differences. Difference symbols are more concentrated at zero for this slowly varying example; this is an example, not a universal decrease of marginal entropy.

Code: `figures-code/chapter-shannon-coding-theory/codage-differences`

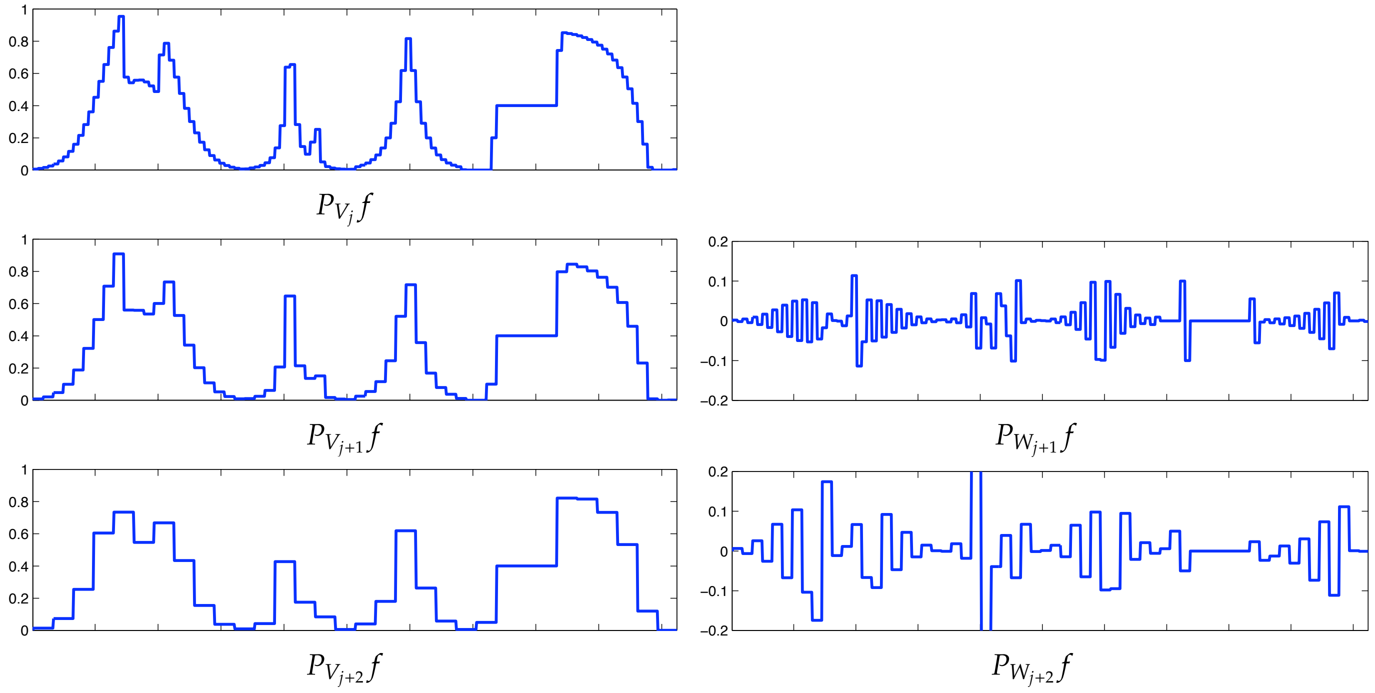
Figure 4.5

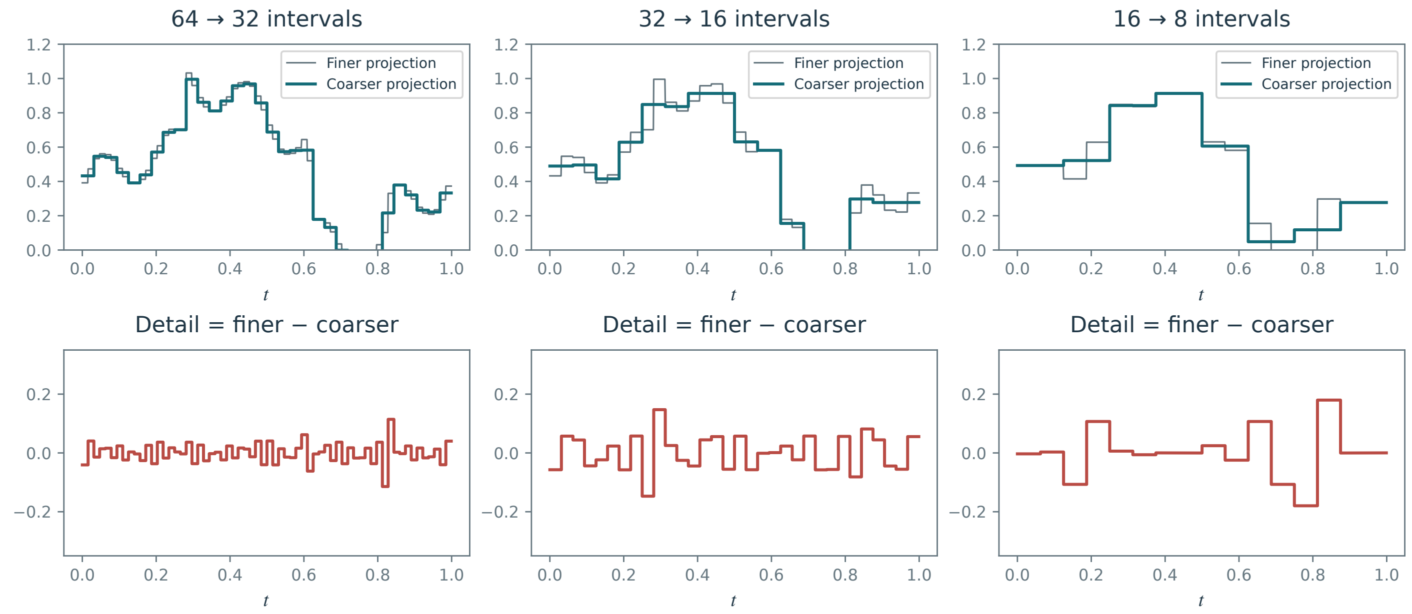
Successive Haar approximations (left) and the extracted details (right). Increasing the relative level from j to $j+1$ and $j+2$ coarsens the approximation.

Wavelets

Current book

Proposed reconstruction





Mathematical and generation notes. Three successive Haar resolution pairs are arranged above their corresponding detail functions, including the additional 64-to-32 interval pair. Each column satisfies finer = coarser + detail and exact coarse/detail orthogonality. The resolution and detail rows use common vertical ranges.

Code: `figures-code/chapter-wavelets/haar-appr-details`

Figure 4.8

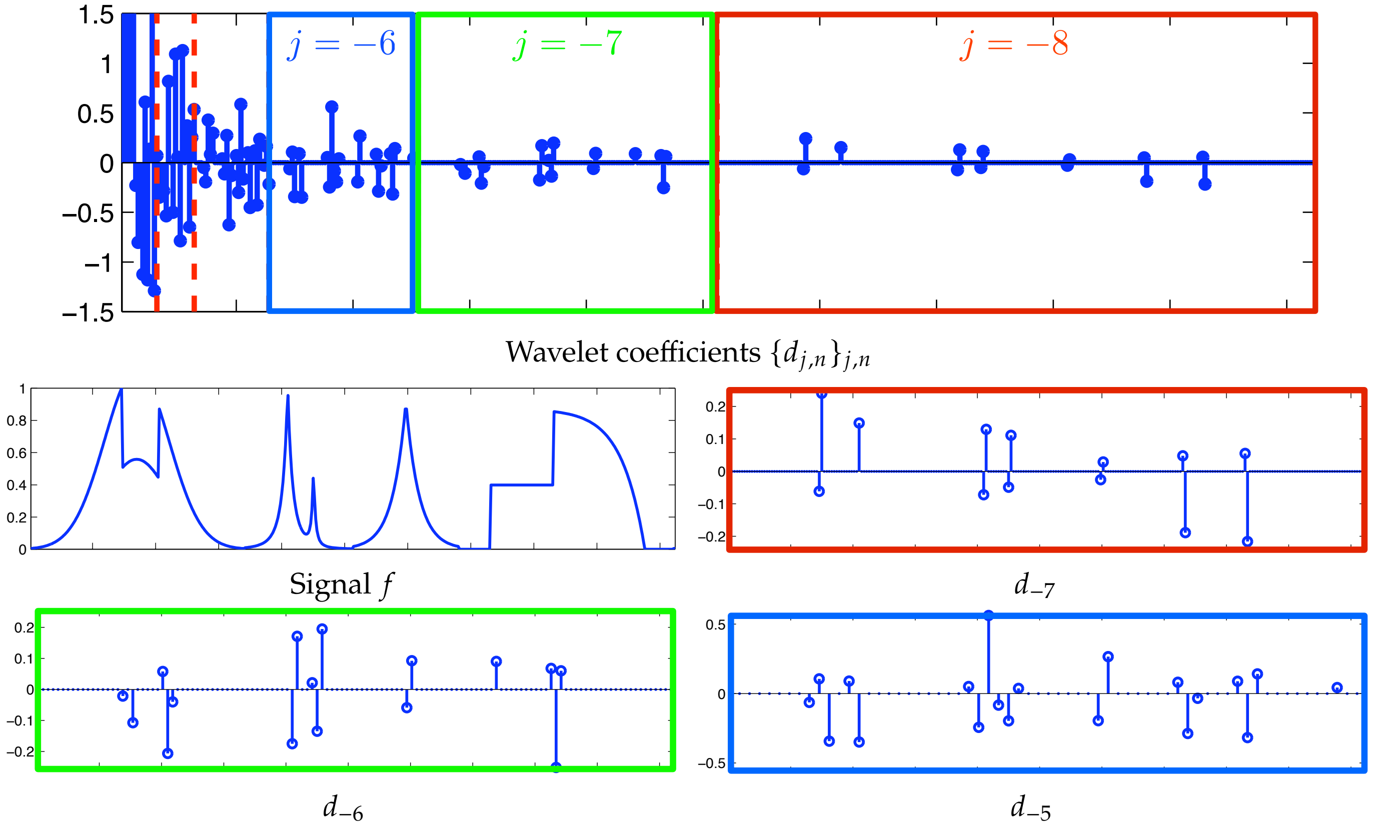
Main book, page 47

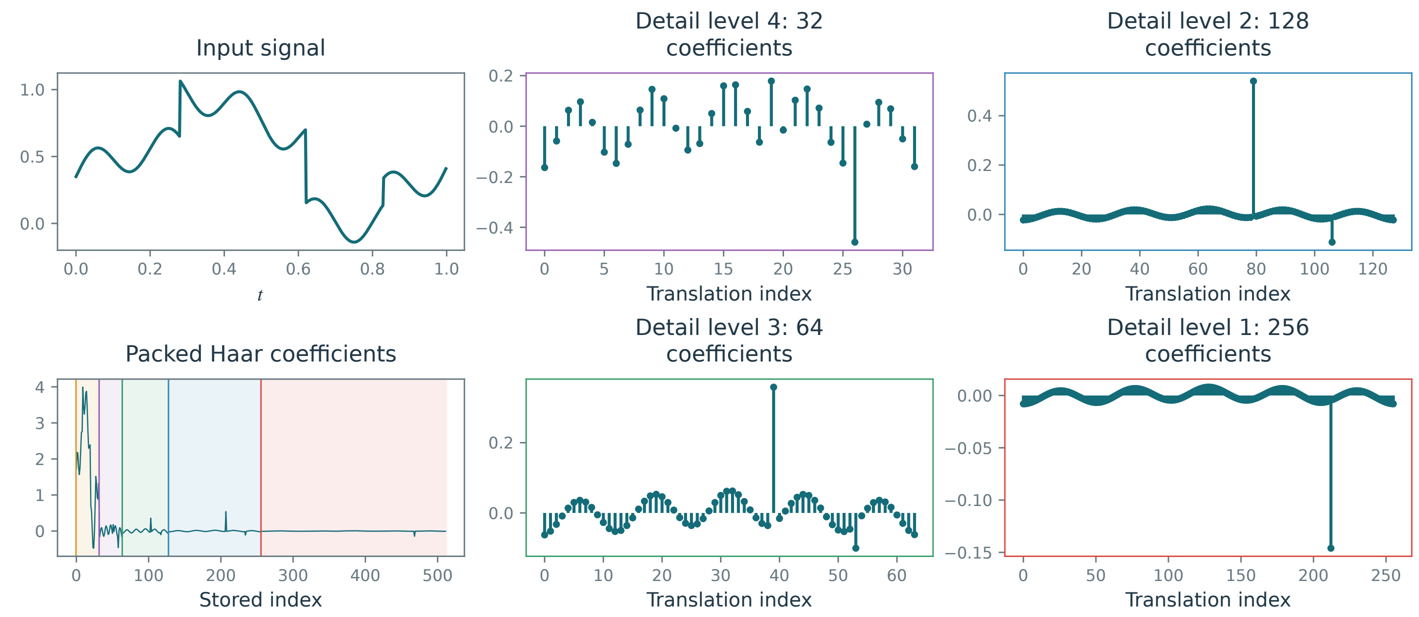
Wavelet coefficients. Top: the complete decomposition. Below: enlarged views of individual scales.

Wavelets

Current book

Proposed reconstruction





Mathematical and generation notes. The first column contains the input and complete packed Haar transform. The next two columns enlarge its four detail bands, with matching scale colors. All coefficients come from the same orthonormal transform; coarse coefficients precede details from coarse to fine.

Code: `figures-code/chapter-wavelets/wavcoef-1d`

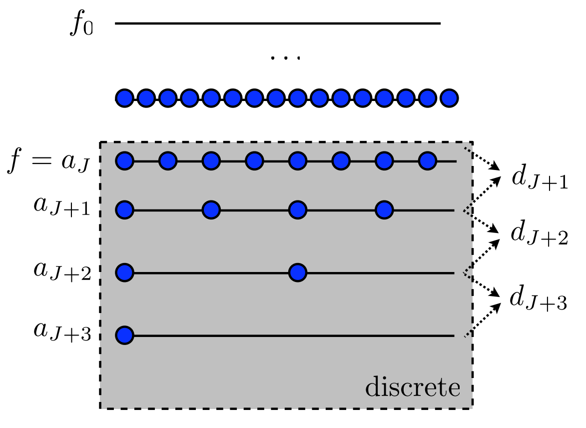
Figure 4.11

Main book, page 49

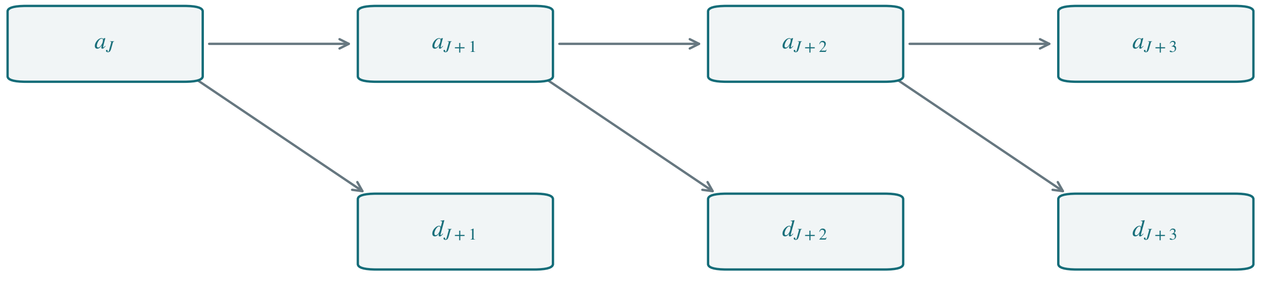
Pyramidal computation of wavelet coefficients.

Wavelets

Current book



Proposed reconstruction



Approximation row a_j; extracted details d_j below

Mathematical and generation notes. Approximation outputs a_j are aligned on the top row; each extracted d_j is directly below the approximation from the same split. Only the approximation is recursively decomposed, and every detail band is stored once.

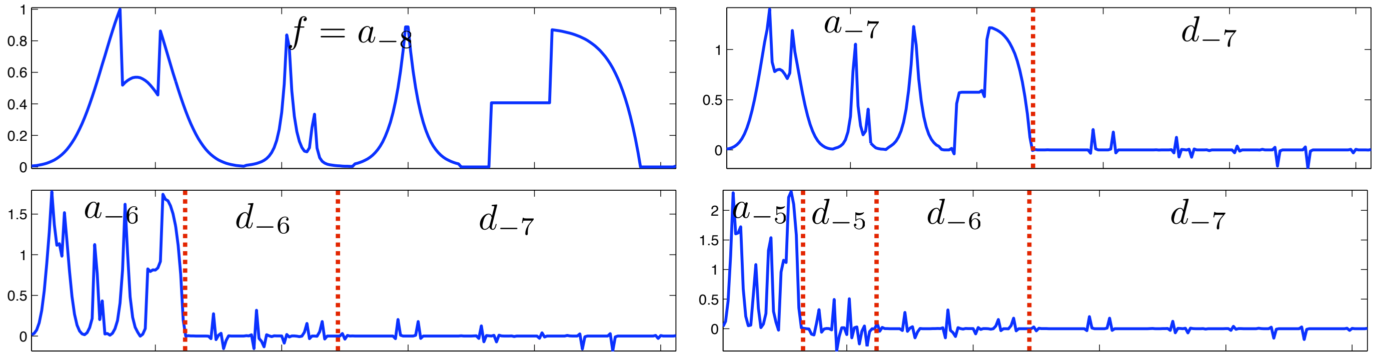
Code: `figures-code/chapter-wavelets/filterbank-pyramidal`

Figure 4.12

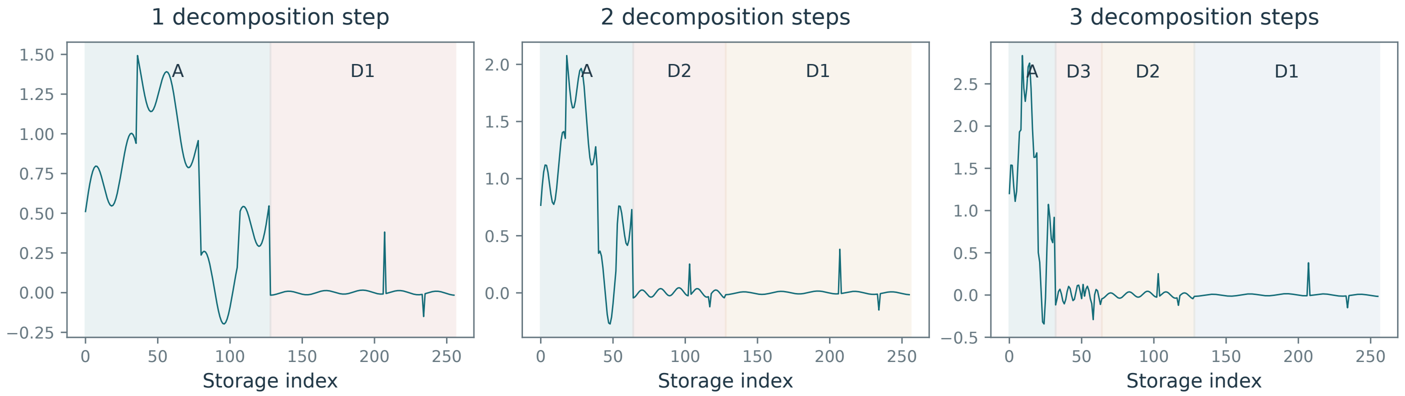
Wavelet decomposition algorithm.

Wavelets

Current book



Proposed reconstruction



Mathematical and generation notes. The storage arrays are recomputed after one, two and three Haar steps. The active approximation occupies the leftmost block and is replaced by a smaller approximation followed by its detail. Earlier details stay in place. Length and energy are preserved at every step.

Code: `figures-code/chapter-wavelets/wavalgo-1d`

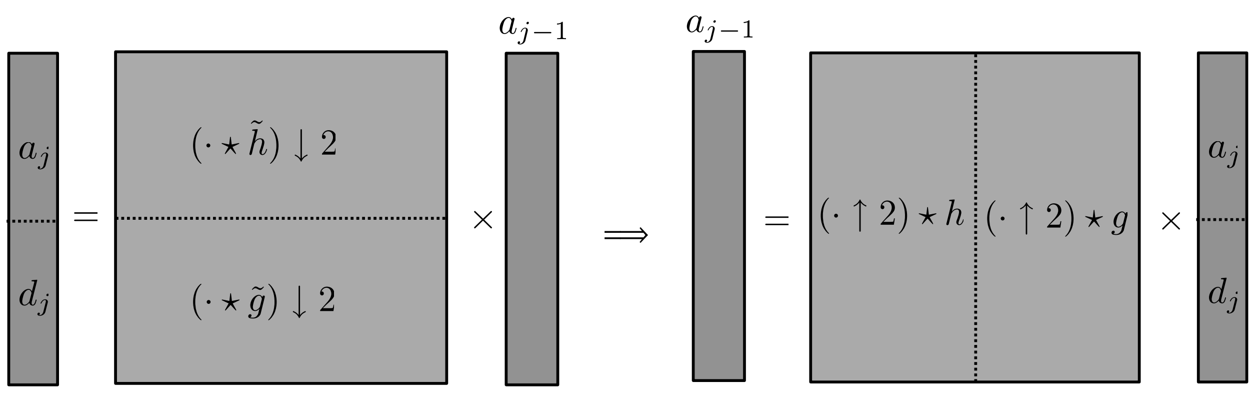
Figure 4.13

Main book, page 51

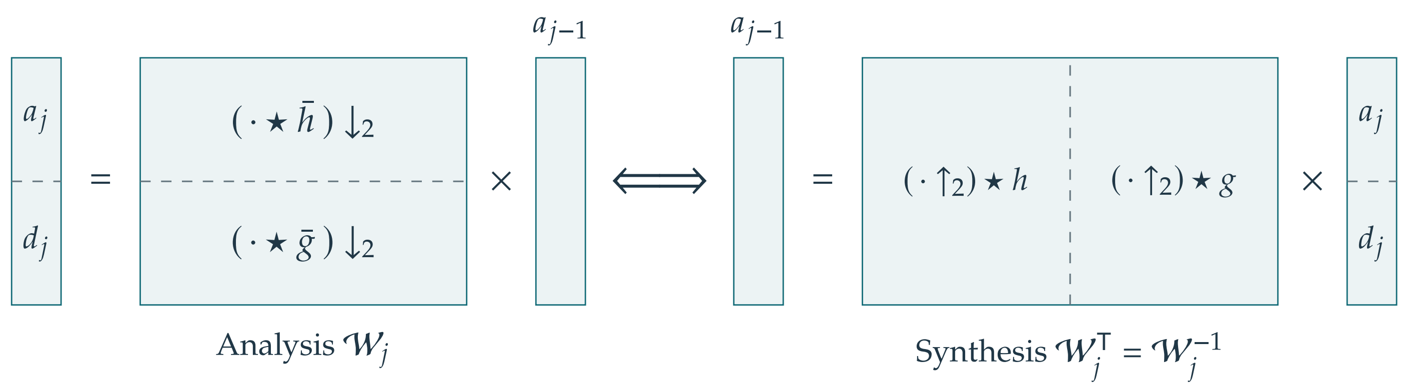
Inverse wavelet transform in matrix form.

Wavelets

Current book



Proposed reconstruction



Mathematical and generation notes. Recreates the original block-operator picture: analysis has two vertically stacked convolution/downsampling blocks; synthesis transposes this arrangement into two horizontally concatenated upsampling/convolution blocks. The stacked vector orders a_j above d_j, and both sides act on the correctly sized a_(j-1). The concrete eight-sample Haar matrix is retained as a separate accepted figure.

Code: `figures-code/chapter-wavelets/inversion-transpose`

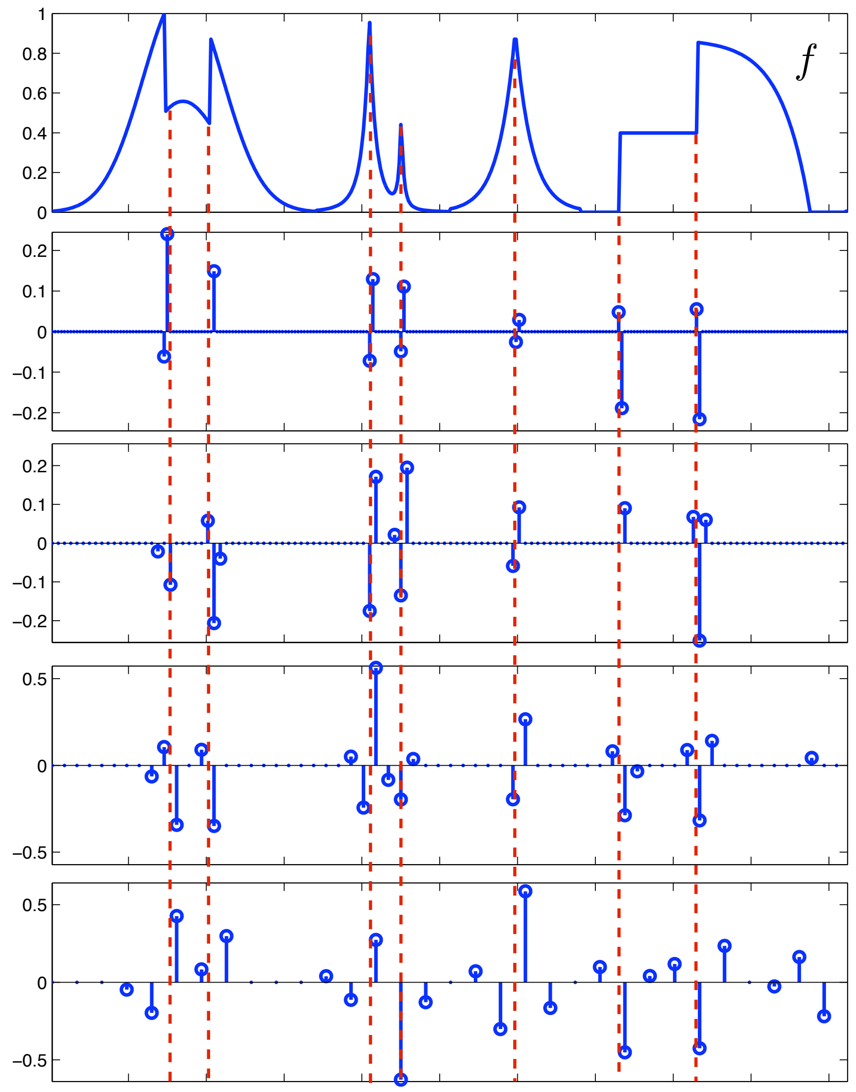
Figure 4.27

Main book, page 62 (previous Figure 4.26)

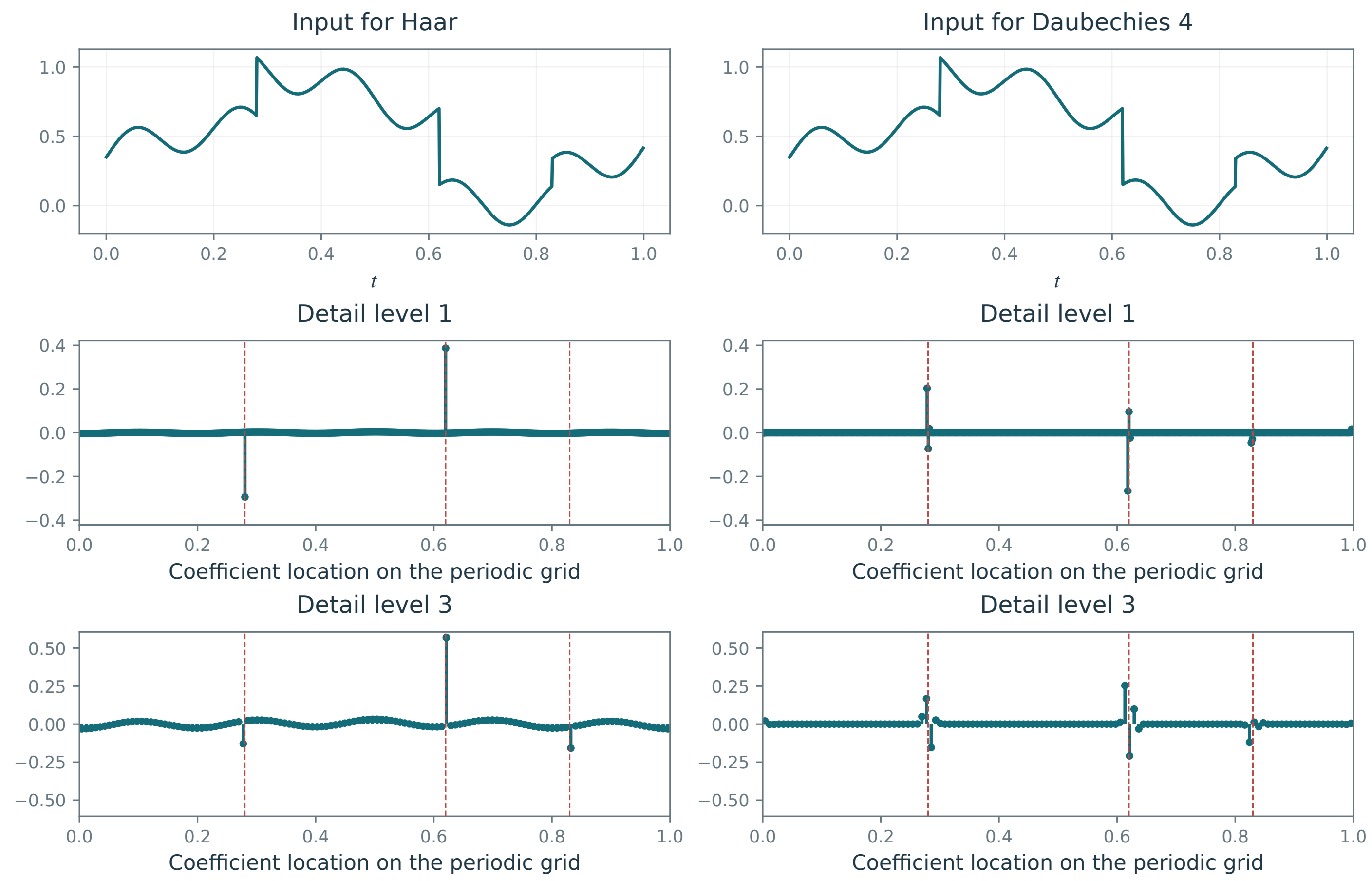
Location of large wavelet coefficients.

Wavelets

Current book



Proposed reconstruction



Mathematical and generation notes. Haar and Daubechies-4 coefficients of the same piecewise-smooth signal are compared at two common detail levels. Dashed lines identify the input jumps; the coefficient grid uses the same periodic packing convention. Daubechies support spans more samples and is not identified with the one-cell Haar support. Rows share vertical ranges.

Code: `figures-code/chapter-wavelets/vanishing-moments-1d`

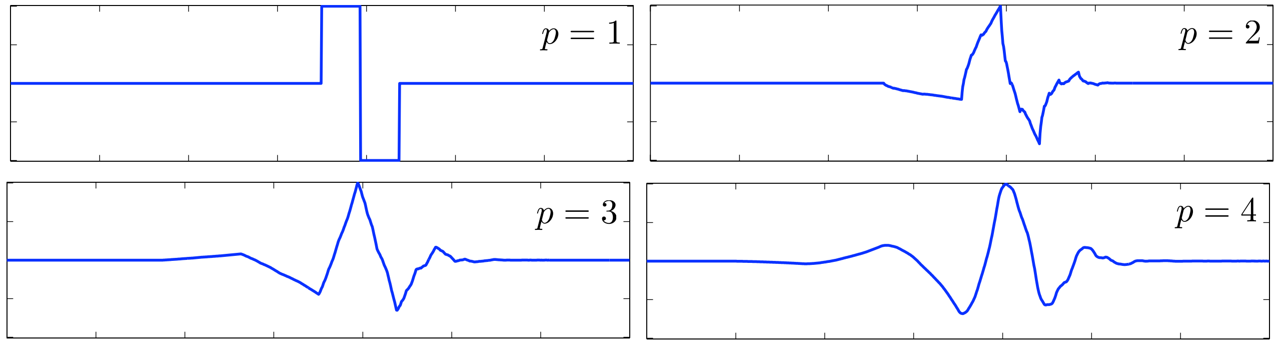
Figure 4.28

Main book, page 63 (previous Figure 4.27)

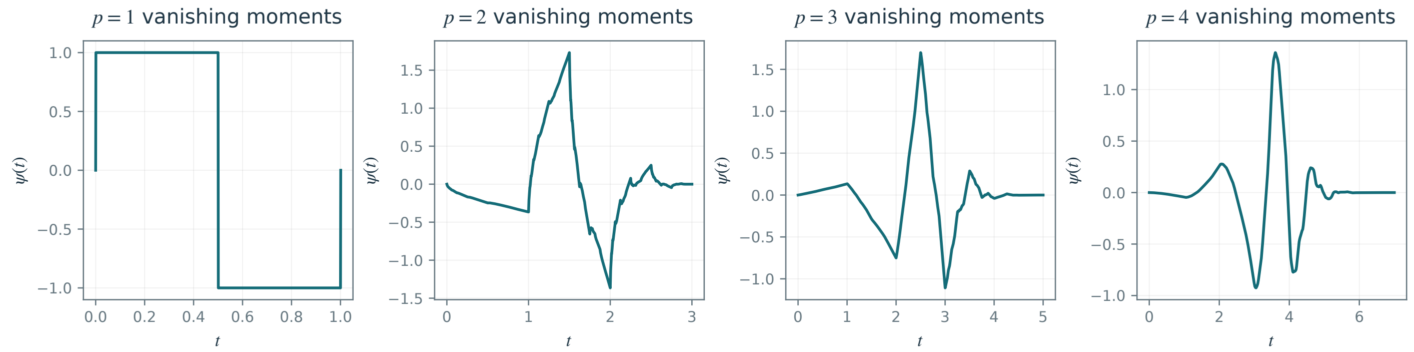
Daubechies mother wavelets ψ with increasing numbers p of vanishing moments.

Wavelets

Current book



Proposed reconstruction



Mathematical and generation notes. The mother wavelets are generated by the Daubechies cascade for p=1,2,3,4. Support length grows as 2p-1. Alternating moments of the scaling filter, which characterize the zero at the Nyquist frequency, are checked through order p-1. Overall wavelet sign follows the library's documented filter convention.

Code: `figures-code/chapter-wavelets/wavelets`

Unnumbered illustration

Unnumbered illustration: embeded spaces 1d

Wavelets

Current book

$$
\begin{tikzcd}
L^2(\mathbb{R}) \arrow[r] & \cdots \arrow[r] \arrow[dr] & V_{j-1} \arrow[r] \arrow[dr] & V_j \arrow[r] \arrow[dr] & V_{j+1} \arrow[r] \arrow[dr] & \cdots \arrow[r] & \{0\} \\
 & & W_{j-1} & W_j & W_{j+1} & &
\end{tikzcd}
$$

Proposed reconstruction

$$
\begin{tikzcd}
\cdots \arrow[r] \arrow[dr] & V_{j-1} \arrow[r] \arrow[dr] & V_j \arrow[r] \arrow[dr] & V_{j+1} \arrow[r] & \cdots \\
 & W_{j-1} & W_j & W_{j+1} &
\end{tikzcd}
$$

$$V_{j-1} = V_j \oplus^{\perp} W_j$$

Mathematical and generation notes. Restores the original two-row cascade: approximation spaces V_j on top and orthogonal detail spaces W_j below. Horizontal arrows are coarse projections; each diagonal extracts the detail complement from the preceding finer space. The identity underneath fixes the chapter convention that larger j is coarser. In two dimensions W_j^(2) contains all three tensor detail orientations.

Code: `figures-code/chapter-wavelets/nested-spaces-1d`

Unnumbered illustration

Unnumbered illustration: embeded spaces 2d

Wavelets

Current book

$$
\begin{tikzcd}
\mathrm{L}^2(\mathbb{R}^2) \arrow[r] & \cdots \arrow[r] \arrow[dr] & V_{j-1} \otimes V_{j-1} \arrow[r] \arrow[dr] & V_j \otimes V_j \arrow[r] \arrow[dr] & V_{j+1} \otimes V_{j+1} \arrow[r] \arrow[dr] & \cdots \arrow[r] & \{0\} \\
 & & W_{j-1}^{(2)} & W_j^{(2)} & W_{j+1}^{(2)} & &
\end{tikzcd}
$$

Proposed reconstruction

$$
\begin{tikzcd}
\cdots \arrow[r] \arrow[dr] & V_{j-1} \otimes V_{j-1} \arrow[r] \arrow[dr] & V_j \otimes V_j \arrow[r] \arrow[dr] & V_{j+1} \otimes V_{j+1} \arrow[r] & \cdots \\
 & W_{j-1}^{(2)} & W_j^{(2)} & W_{j+1}^{(2)} &
\end{tikzcd}
$$

$$V_{j-1} \otimes V_{j-1} = (V_j \otimes V_j) \oplus^{\perp} W_j^{(2)}$$

Mathematical and generation notes. Restores the original two-row cascade: approximation spaces V_j on top and orthogonal detail spaces W_j below. Horizontal arrows are coarse projections; each diagonal extracts the detail complement from the preceding finer space. The identity underneath fixes the chapter convention that larger j is coarser. In two dimensions W_j^(2) contains all three tensor detail orientations.

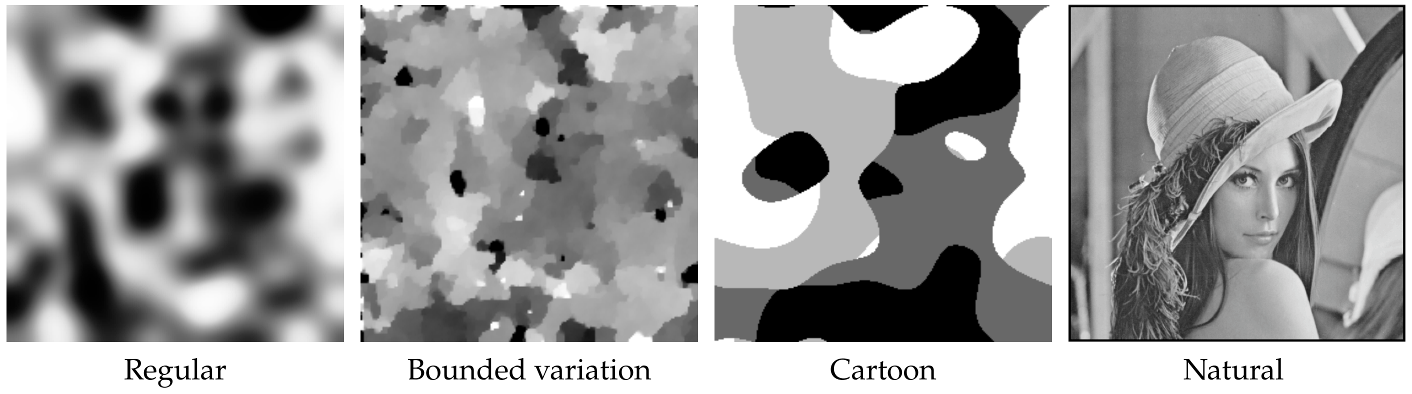
Code: `figures-code/chapter-wavelets/nested-spaces-2d`

Figure 5.4

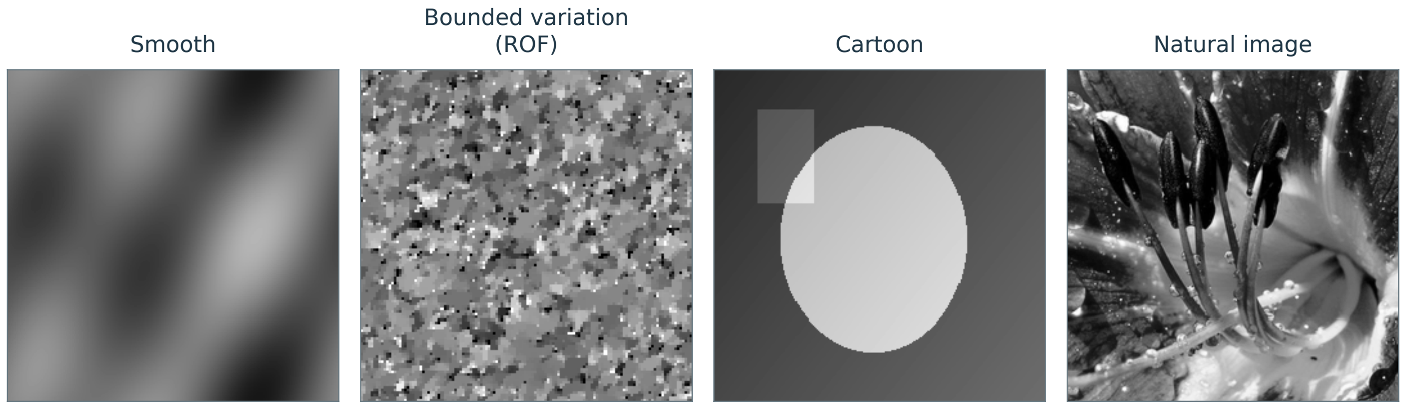
Examples of image models.

Linear and Nonlinear Approximation

Current book



Proposed reconstruction



Mathematical and generation notes. The BV panel is obtained by solving the ROF total-variation denoising problem on a seeded pure white-noise image. An affine rescaling makes its piecewise-flat structure visible. The classes overlap: smooth and cartoon functions also belong to BV.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/approximation-class`

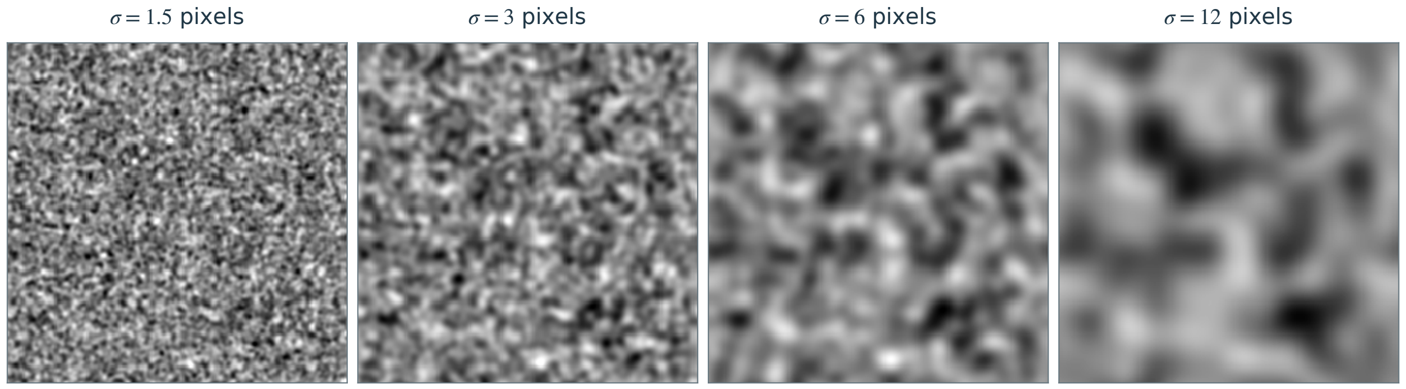
Figure 5.5

Images with increasing Sobolev norm.

Linear and Nonlinear Approximation

Current book

Proposed reconstruction



Mathematical and generation notes. One seeded pure white-noise image is progressively convolved with wider periodic Gaussian kernels. For comparison, each result is affinely normalized to the same mean and standard deviation. The measured gradient energy decreases with smoothing even after this contrast normalization.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/signal-model-sobolev`

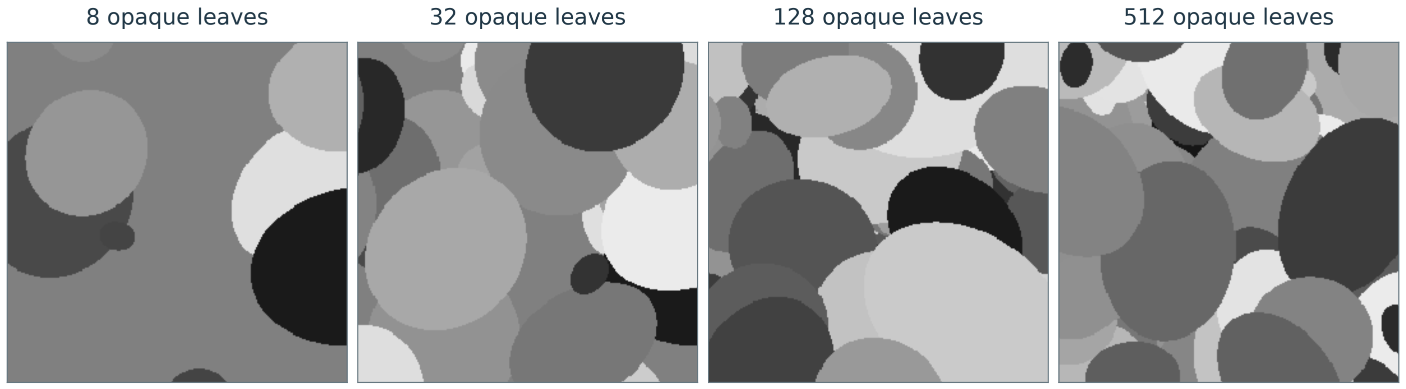
Figure 5.6

Cartoon images with increasing total variation.

Linear and Nonlinear Approximation

Current book

Proposed reconstruction



Mathematical and generation notes. A progressive dead-leaves model overlays opaque ellipses with independently sampled positions, sizes, orientations and gray values. Later leaves occlude earlier ones. Every snapshot is piecewise constant with a finite union of smooth boundary arcs; intersections and occlusions create corners. Total variation is measured, not assumed monotone under occlusion.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/image-model-cartoon`

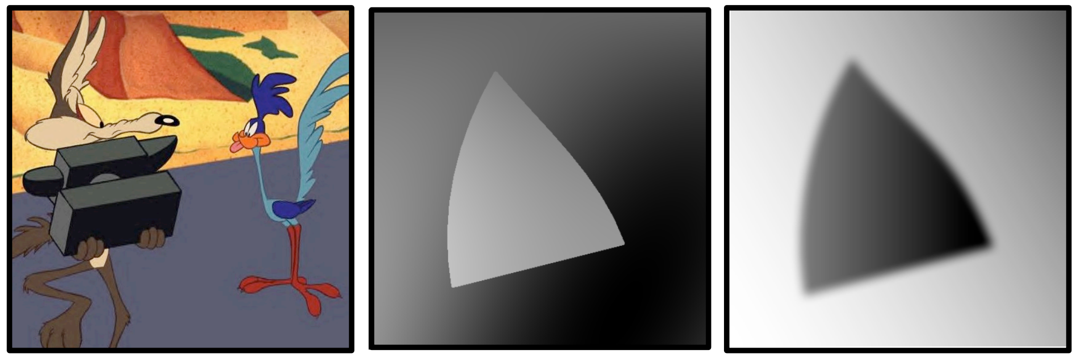
Figure 5.7

Main book, page 69

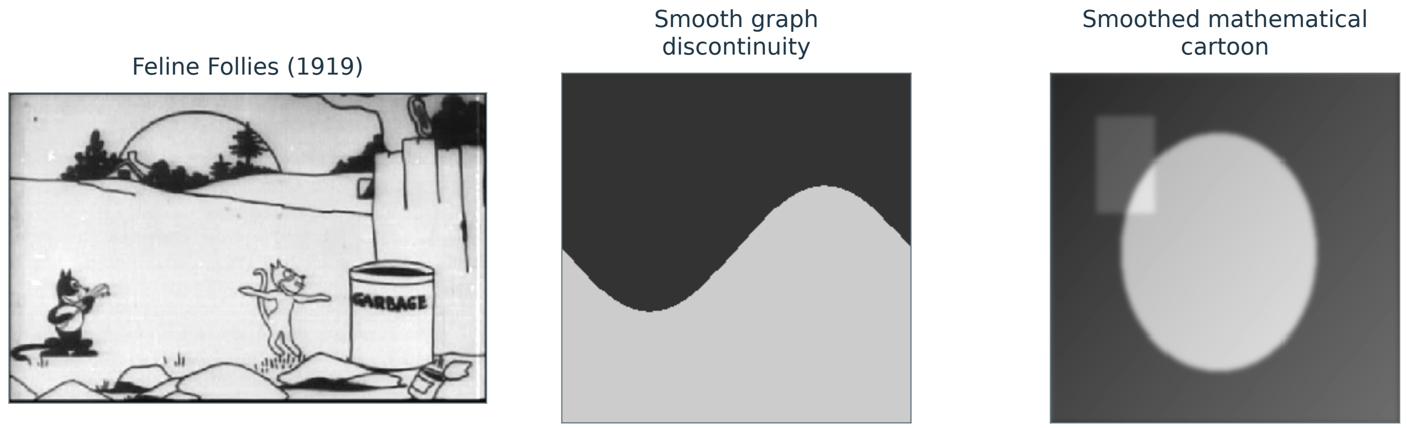
Examples of cartoon images: sharp discontinuities (left and center) and blurred edges (right).

Linear and Nonlinear Approximation

Current book



Proposed reconstruction



Mathematical and generation notes. The first panel is a real grayscale animation still from Feline Follies (1919), attributed by Wikimedia Commons to Pat Sullivan and listed there as public domain. Source: https://commons.wikimedia.org/wiki/File:Felix_1919.jpg . The other panels illustrate a mathematically defined jump along a smooth graph and an actual Gaussian-smoothed cartoon. An animation drawing and the mathematical cartoon class are distinguished.

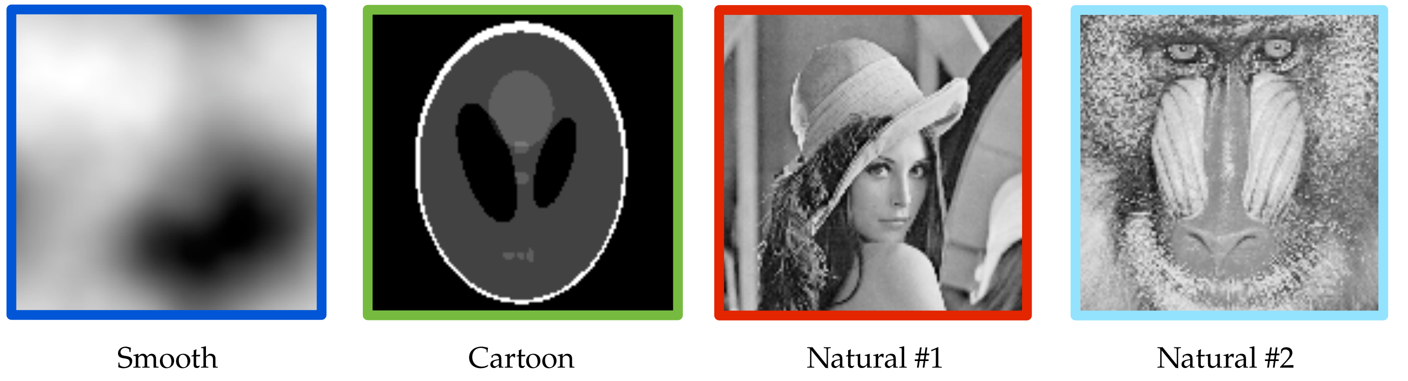
Code: `figures-code/chapter-linear-and-nonlinear-approximation/sample-cartoon`

Figure 5.8

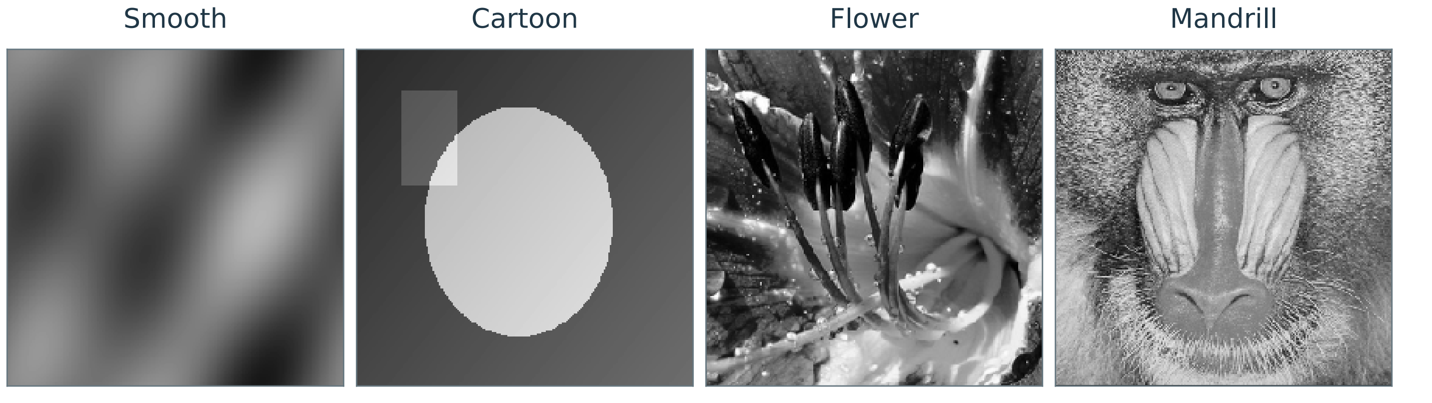
Test images with different types of structure.

Linear and Nonlinear Approximation

Current book



Proposed reconstruction



Mathematical and generation notes. These four precisely defined test images are reused unchanged in the next approximation-error figure: smooth, cartoon, the supplied flower and the author-requested mandrill.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/approx-speed-1`

Figure 5.10

Main book, page 71

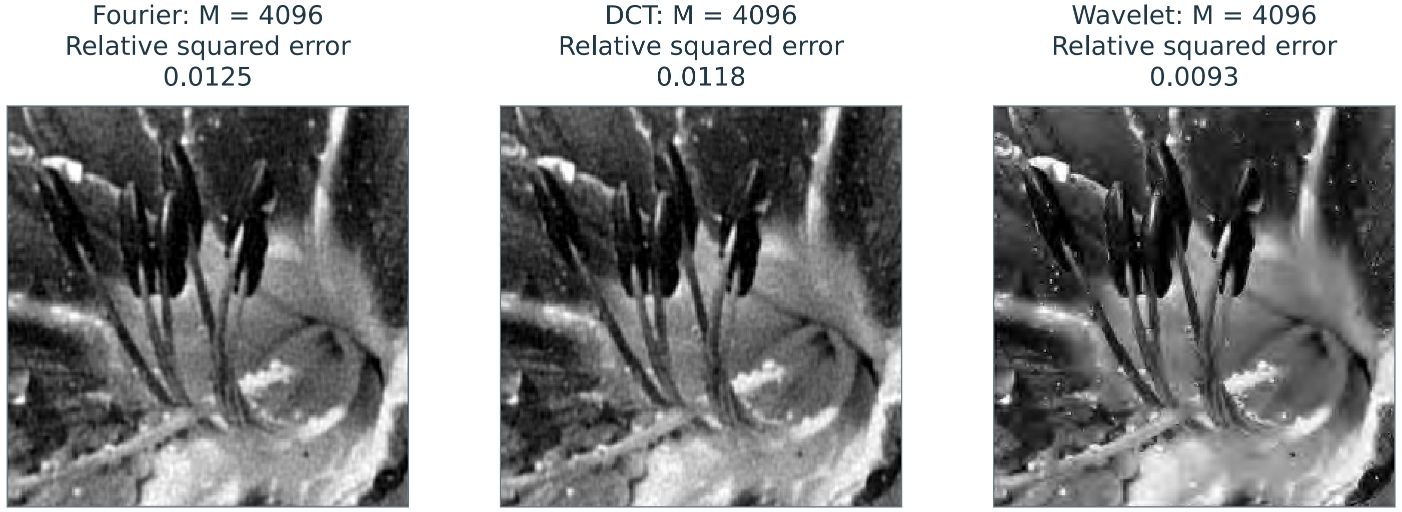
Comparison of approximation errors for different bases using the same number $M = N/50$ of coefficients.

Linear and Nonlinear Approximation

Current book

Fourier SNR=17.1dB | Cosine SNR=17.5dB | Local DCT SNR=18.4dB | Wavelets SNR=19.3dB

Proposed reconstruction



Mathematical and generation notes. All three orthonormal representations retain exactly 4096 real coefficients from the same 256 by 256 flower image. Fourier counts individual real sine/cosine atoms; wavelets use four periodic db4 levels. At this larger common budget, wavelets have the lowest measured error. This observation concerns this image and budget, not a universal ordering of bases.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/approx-effi-1`

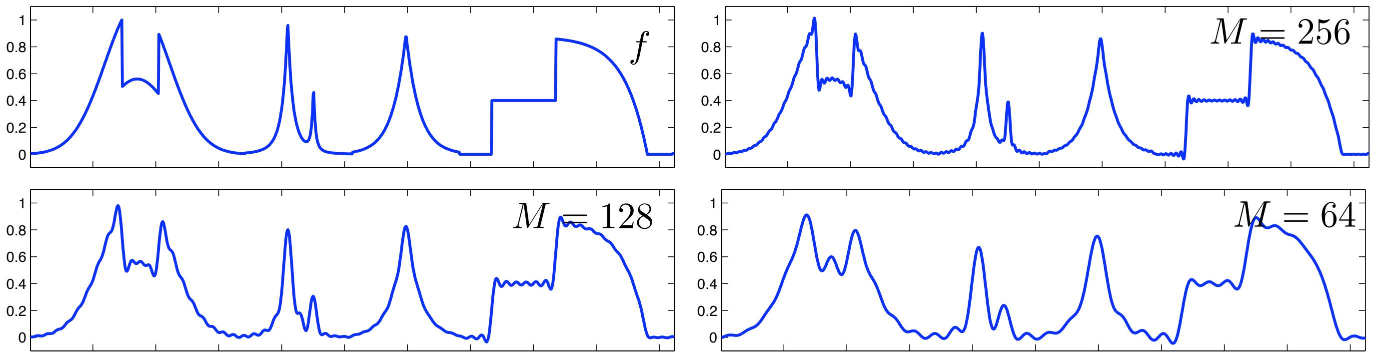
Figure 5.13

Main book, page 73

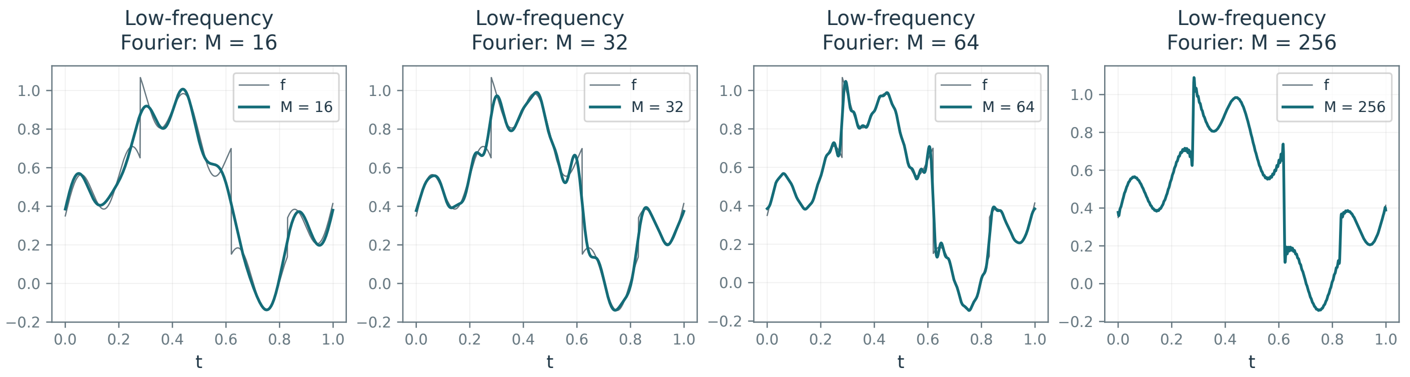
Fourier approximation of a signal.

Linear and Nonlinear Approximation

Current book



Proposed reconstruction



Mathematical and generation notes. Low-frequency real Fourier projections of the same piecewise-smooth periodic signal show Gibbs oscillations near jumps. Counts include the constant and individual sine/cosine atoms; ties in frequency are resolved deterministically.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/fourier-approximation-1d`

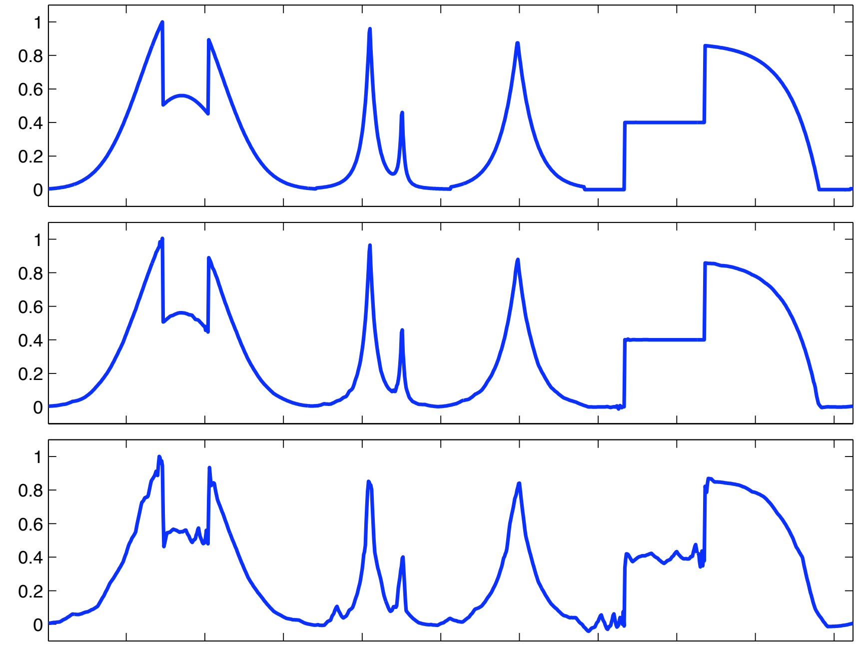
Figure 5.17

Main book, page 77

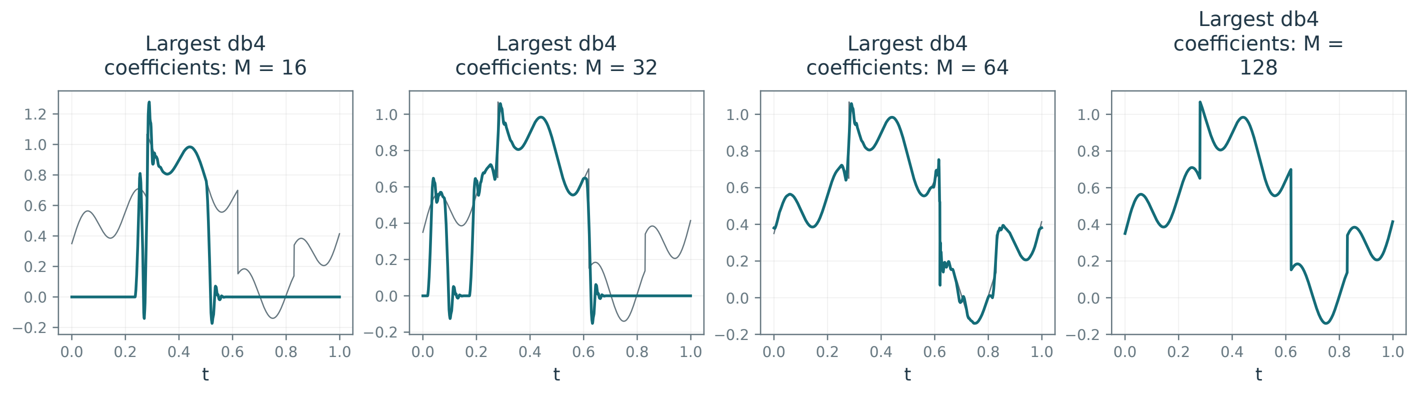
1-D wavelet approximation.

Linear and Nonlinear Approximation

Current book



Proposed reconstruction



Mathematical and generation notes. Nonlinear orthonormal wavelet approximations retain the largest 16,32,64,128 coefficients of the same 1024-sample signal. Fine details survive primarily near jumps.

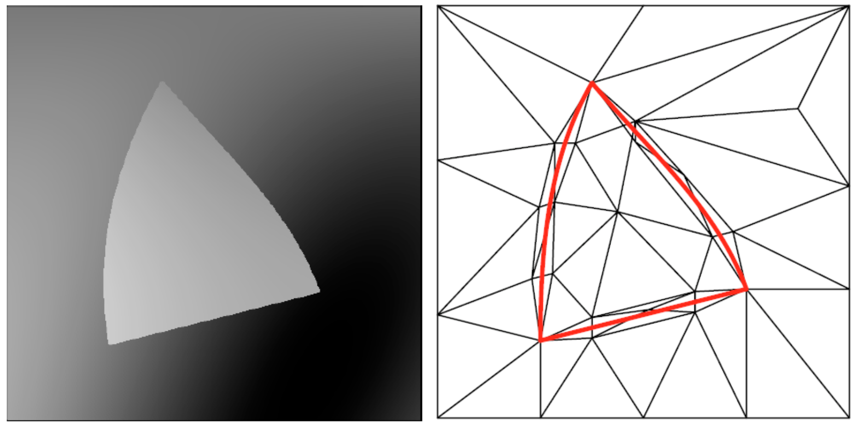
Code: `figures-code/chapter-linear-and-nonlinear-approximation/wav-approx-1d`

Figure 5.20

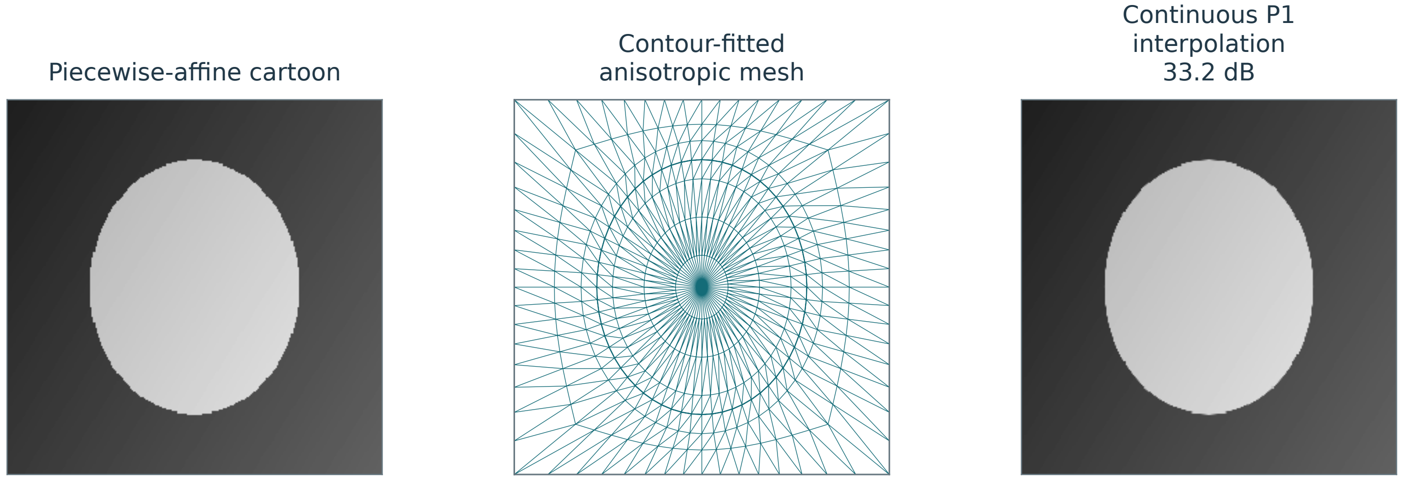
Left: cartoon image, right: adaptive triangulation.

Linear and Nonlinear Approximation

Current book



Proposed reconstruction



Mathematical and generation notes. A conforming triangulation is explicitly fitted to the known elliptical discontinuity. Two close contour rings form elongated tangential elements across the jump; coarser rings fill smooth regions. The third panel is the actual continuous piecewise-linear nodal interpolant. The mesh uses the known synthetic contour, not a claim of automatic optimal segmentation.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/triangulation-image`

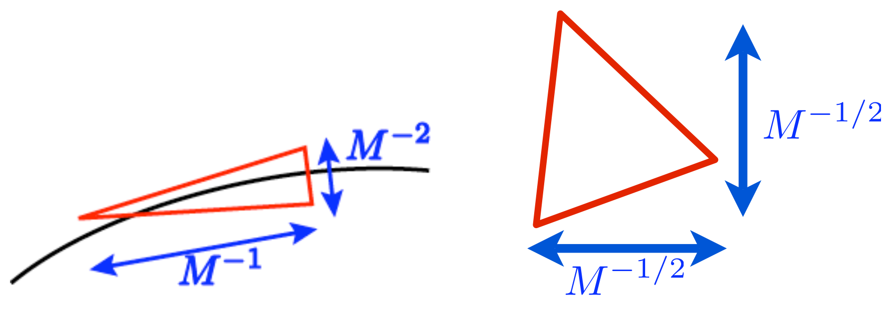
Figure 5.21

Main book, page 80

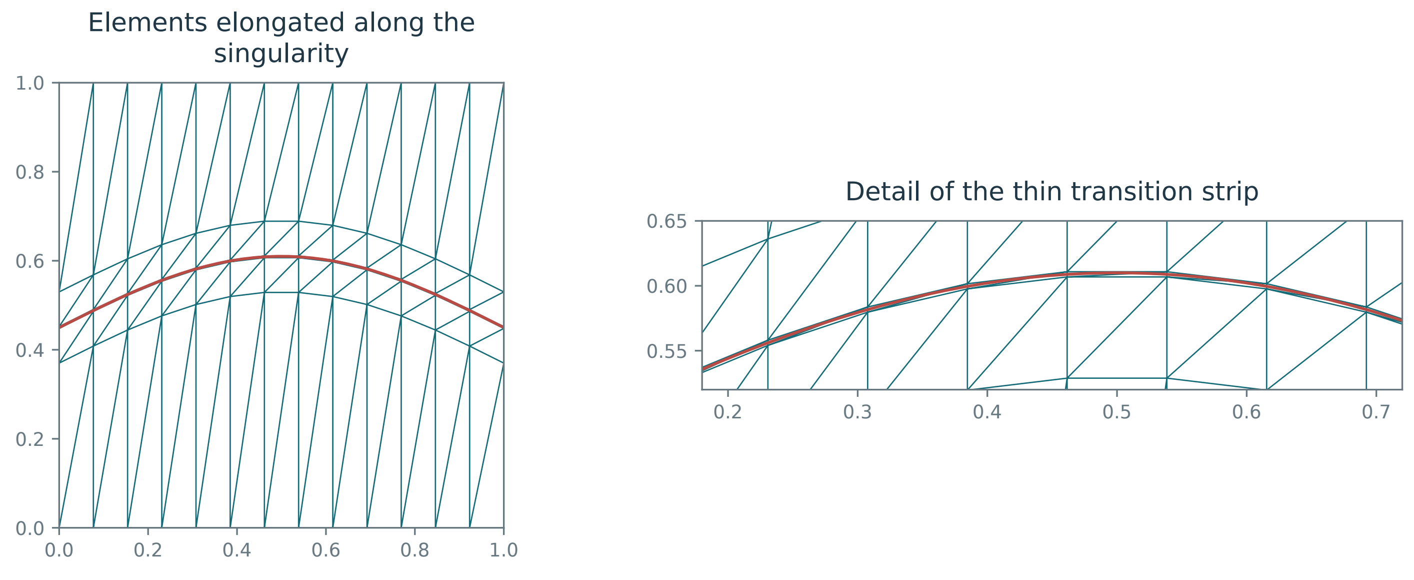
Anisotropic triangles near an edge (left) and approximately isotropic triangles away from edges (right).

Linear and Nonlinear Approximation

Current book



Proposed reconstruction



Mathematical and generation notes. The red curve is the singularity. Explicit strip connectivity creates long tangential edges and very short normal edges on either side of it, rather than an unconstrained Delaunay triangulation. The second panel magnifies the same mesh with equal axis units; it is not a distorted plotting aspect.

Code: `figures-code/chapter-linear-and-nonlinear-approximation/triangulation-anisoiso`

Figure 5.22

Main book, page 80

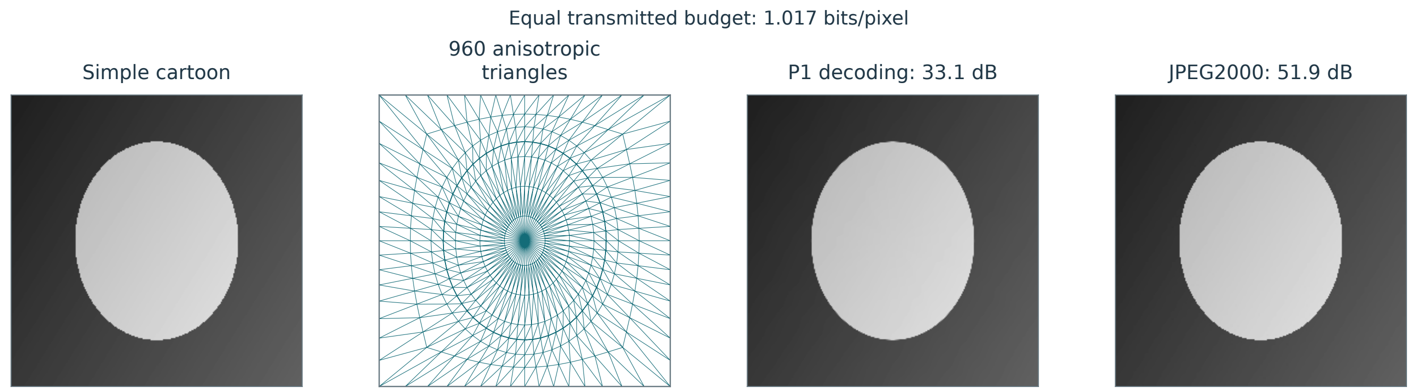
Comparison of adaptive triangulation and JPEG-2000, with the same number of bits.

Linear and Nonlinear Approximation

Current book

Proposed reconstruction

Adapted triangulation $\tilde{f}_M$ JPEG-2000



Mathematical and generation notes. Uses a simpler piecewise-affine cartoon and 960 explicitly connected triangles fitted to its jump. The continuous P1 reconstruction is decoded from a real mesh stream containing a 10-byte header, uint16 coordinates, uint8 nodal values and uint16 triangle indices. The mesh and JPEG2000 streams have equal transmitted byte lengths, including all connectivity, headers and padding. No optimal entropy-coding claim is made.

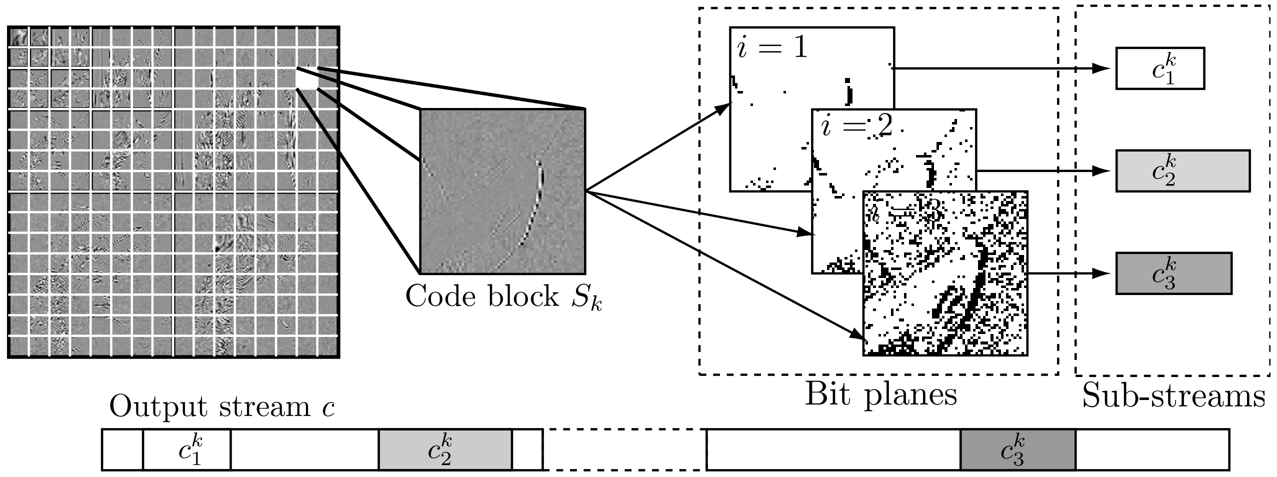
Code: `figures-code/chapter-linear-and-nonlinear-approximation/triangulation-peppers`

Figure 6.4

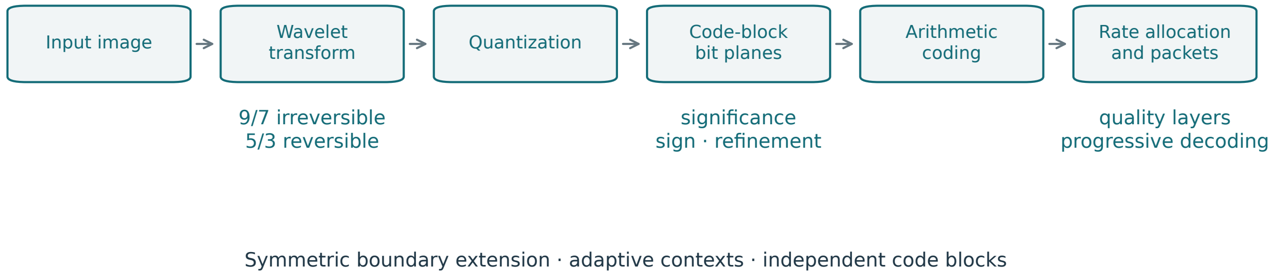
JPEG-2000 coding architecture.

Compression

Current book



Proposed reconstruction



Mathematical and generation notes. The architecture separates transform, quantization, embedded code-block coding, context-adaptive arithmetic coding, and rate/layer organization. It distinguishes irreversible CDF 9/7 coding from reversible integer 5/3 coding. The diagram describes the stages in the chapter without implying that arbitrary byte truncation remains decodable.

Code: `figures-code/chapter-compression/jpeg-2k-architecture`

Figure 6.5

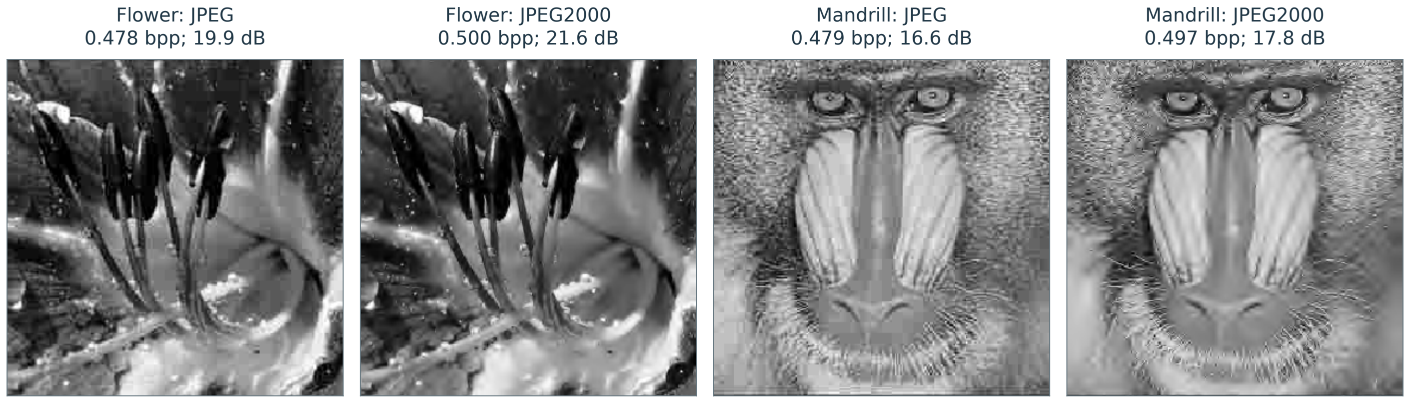
Main book, page 89

Comparison of JPEG (top) and JPEG-2000 (bottom) coding.

Compression

Current book

Proposed reconstruction



Mathematical and generation notes. One row compares JPEG and JPEG-2000 on the flower and a different second image, the author-requested mandrill. Both encoders share the same uncompressed reference and a 0.5 bits/pixel target. Labels count actual stream bytes including headers; panels are decoded from the saved streams.

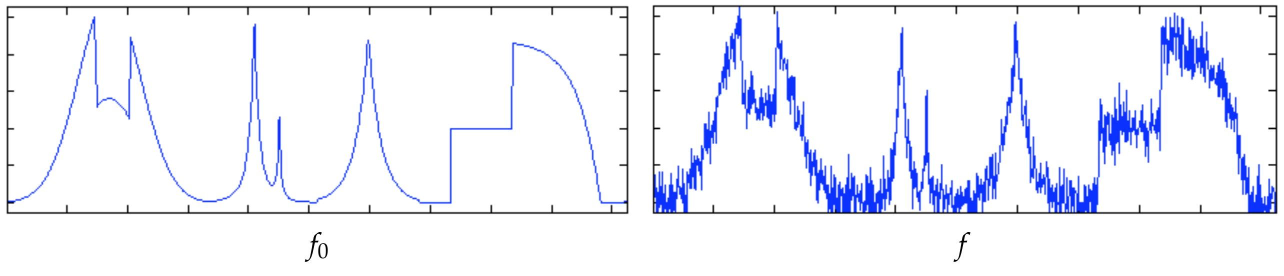
Code: `figures-code/chapter-compression/compression-jpeg`

Figure 7.3

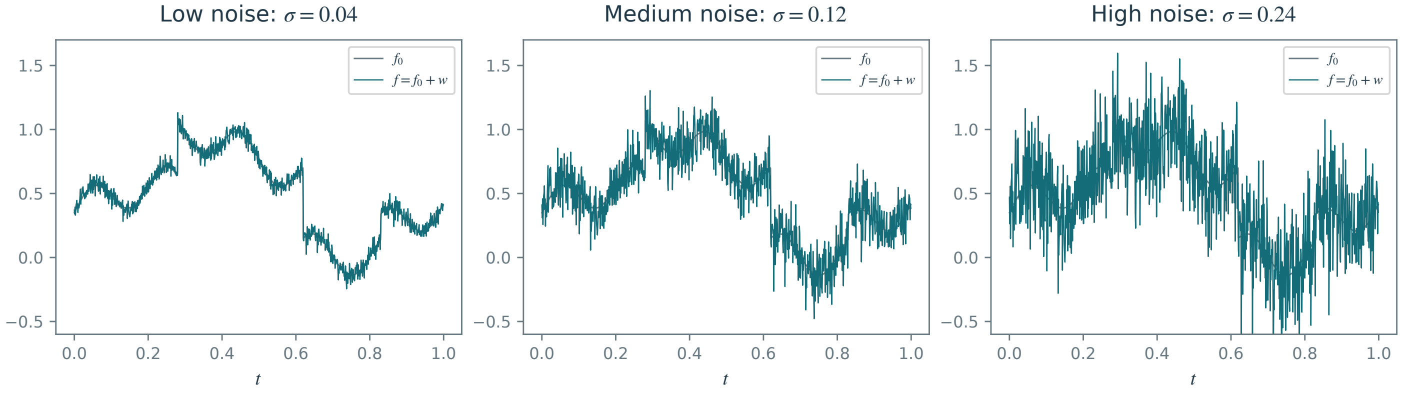
1-D additive noise example.

Denoising

Current book



Proposed reconstruction



Mathematical and generation notes. Three noise levels use the same clean signal and the same standard Gaussian realization, scaled by sigma. This isolates noise amplitude from changes in the underlying sample. All panels share their vertical range.

Code: `figures-code/chapter-denoising/noise-example-1d`

Figure 7.4

Main book, page 92

2-D additive noise example.

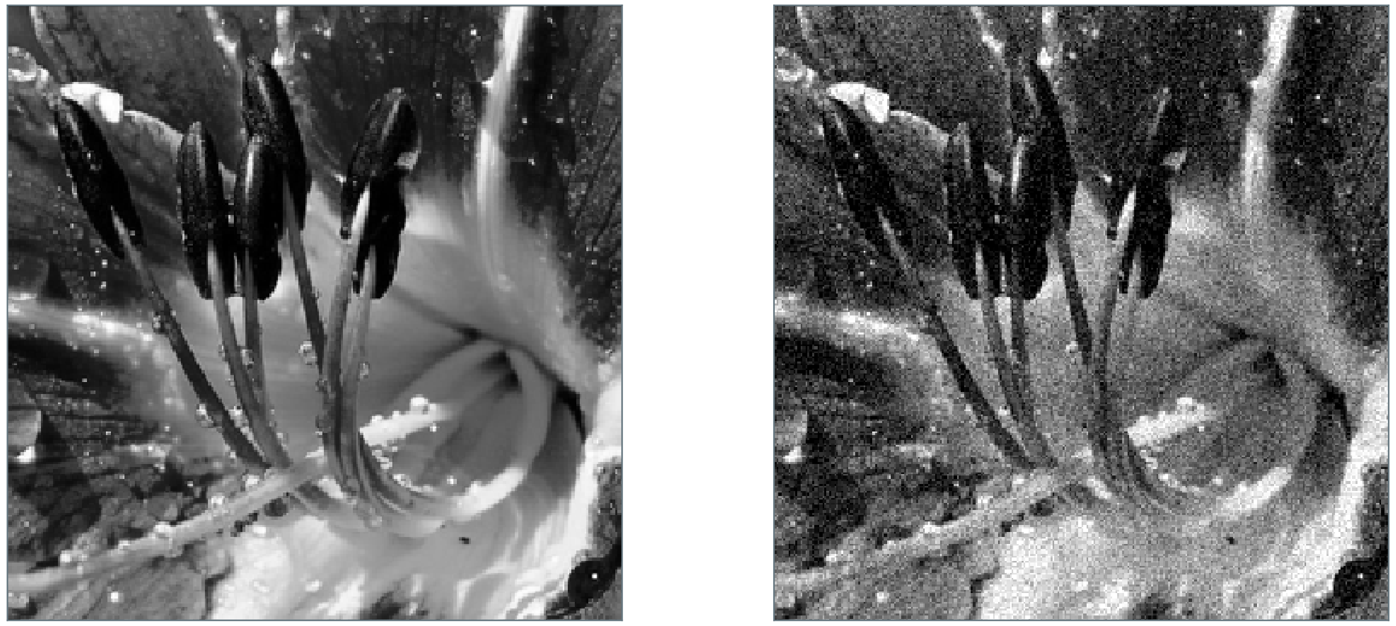
Denoising

Current book

f_0 $\quad$ $y \sim Y = f_0 + w$

Proposed reconstruction

f_0 $\quad$ $f = f_0 + w, \quad w \sim \mathcal{N}(0, \sigma^2 I)$



Mathematical and generation notes. A single fixed white-Gaussian realization is added to the supplied flower luminance. Both images use the range [0,1]; clipping is for display only. This same clean/noisy pair is reused in the subsequent thresholding comparisons.

Code: `figures-code/chapter-denoising/noise-example-2d`

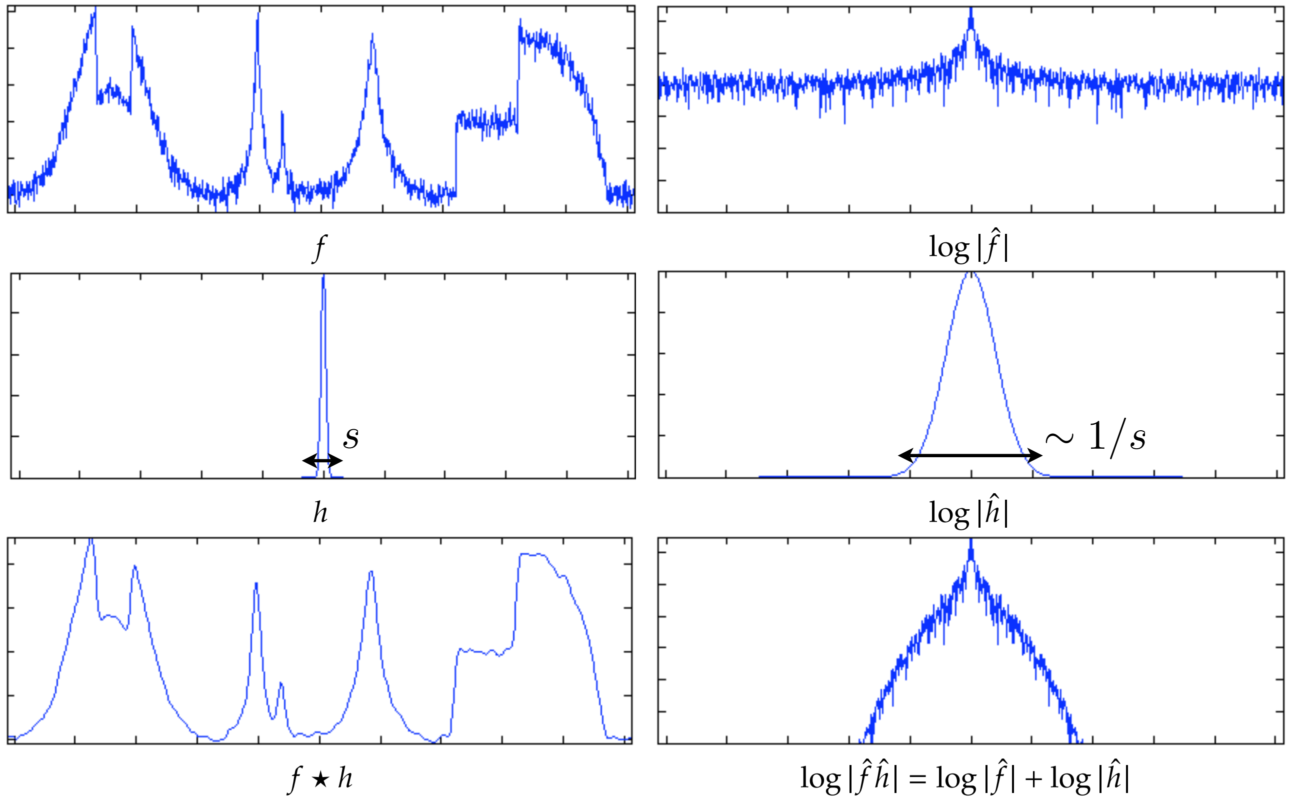
Figure 7.6

Main book, page 93

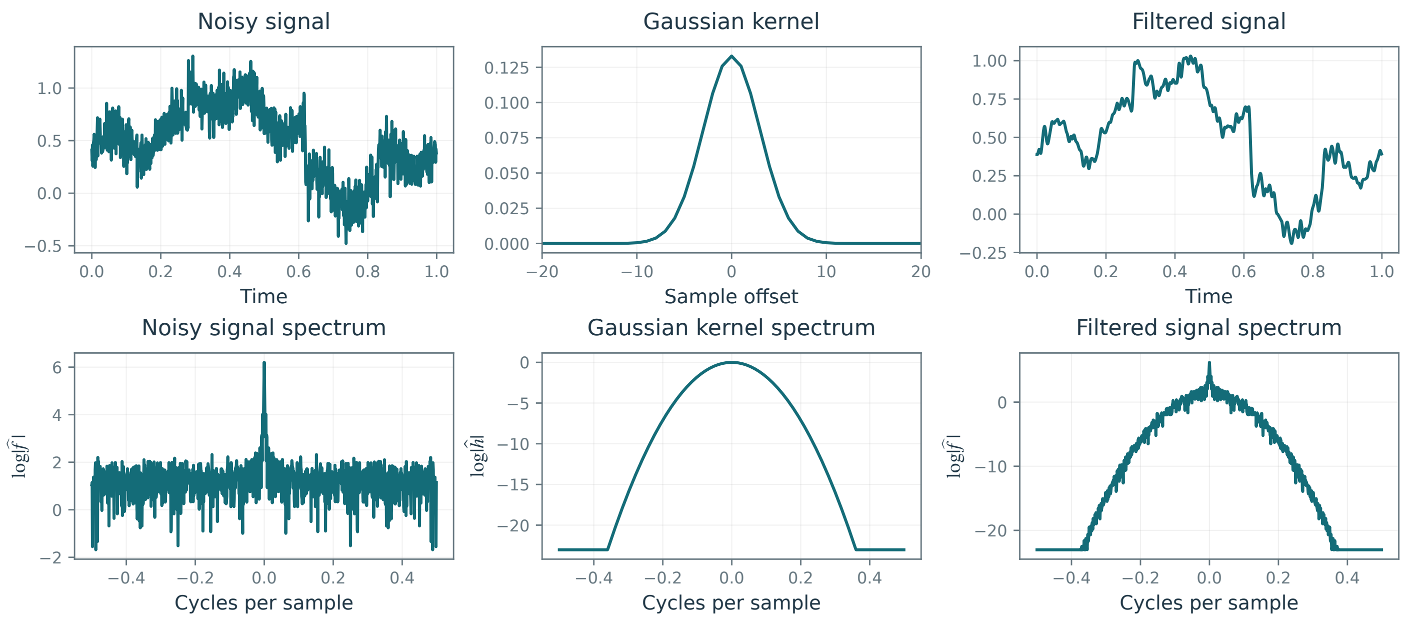
Filtering in the spatial domain (left) and the Fourier domain (right).

Denoising

Current book



Proposed reconstruction



Mathematical and generation notes. The top row contains signal, kernel and filtered signal; the bottom row contains their corresponding Fourier spectra. All six panels belong to one cyclic convolution g=y*h, G=YH. The Gaussian has unit discrete mass; the plotting floor changes only logarithmic display.

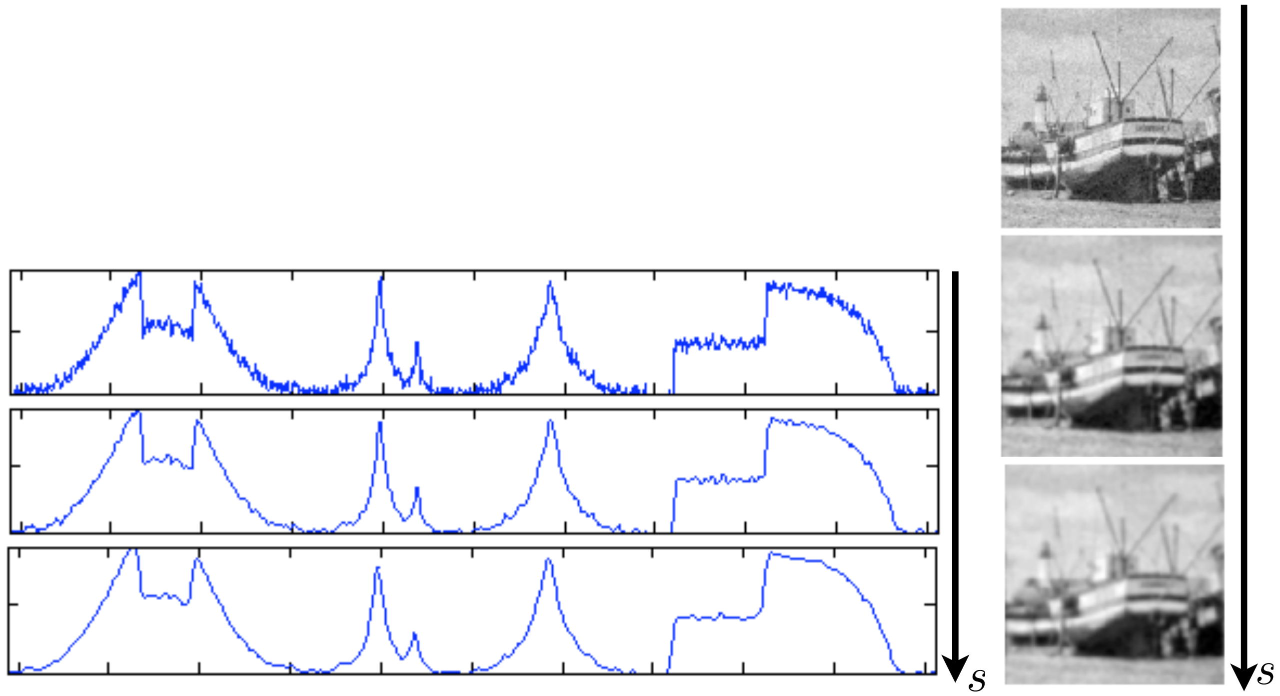
Code: `figures-code/chapter-denoising/filtering-denoised`

Figure 7.7

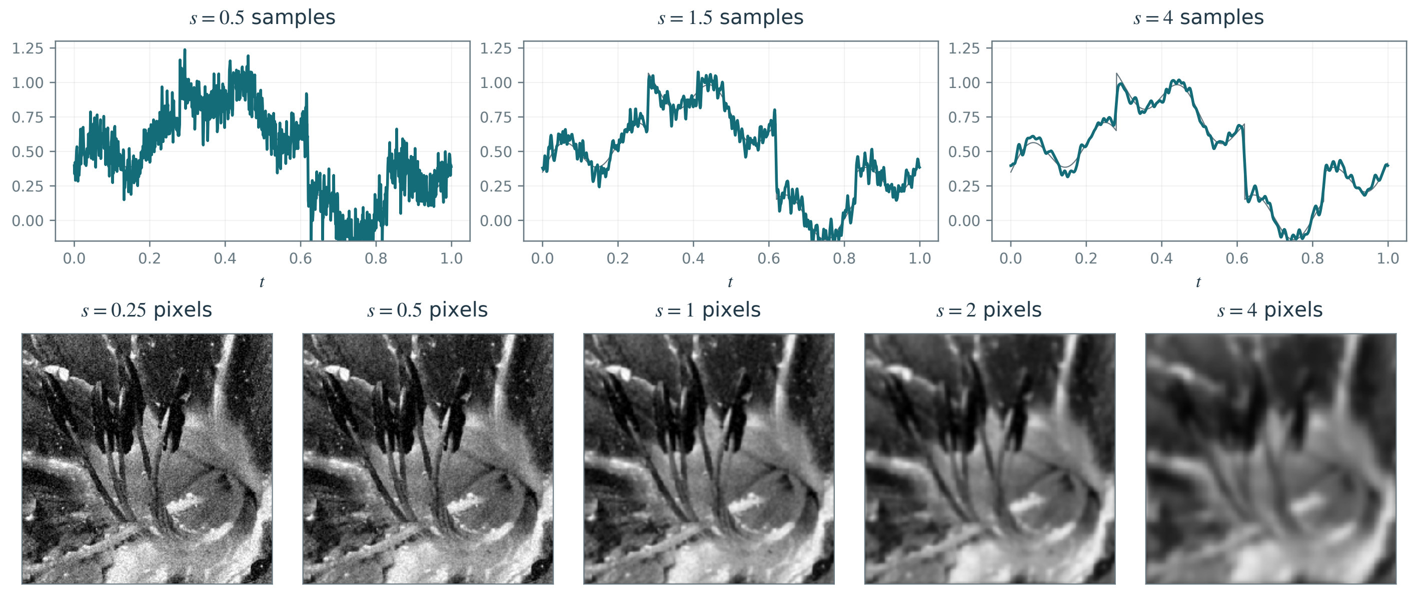
Denoising using a filter of increasing width s.

Denoising

Current book



Proposed reconstruction



Mathematical and generation notes. Three signal views form the first row and five image views the second. Each row reuses a single noisy observation while increasing the standard deviation of the unit-mass periodic Gaussian filter.

Code: `figures-code/chapter-denoising/filtering-progression`

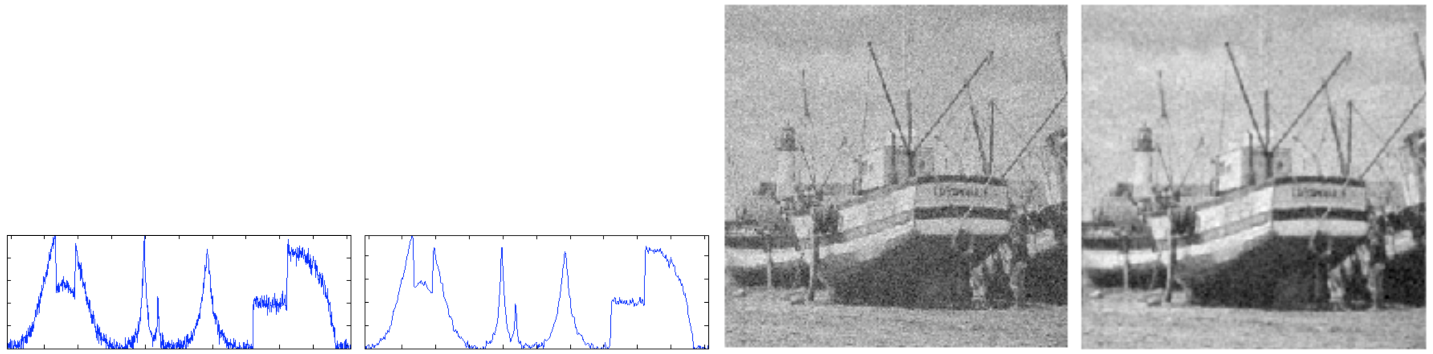
Figure 7.9

Main book, page 95; previous Figures 7.9–7.10

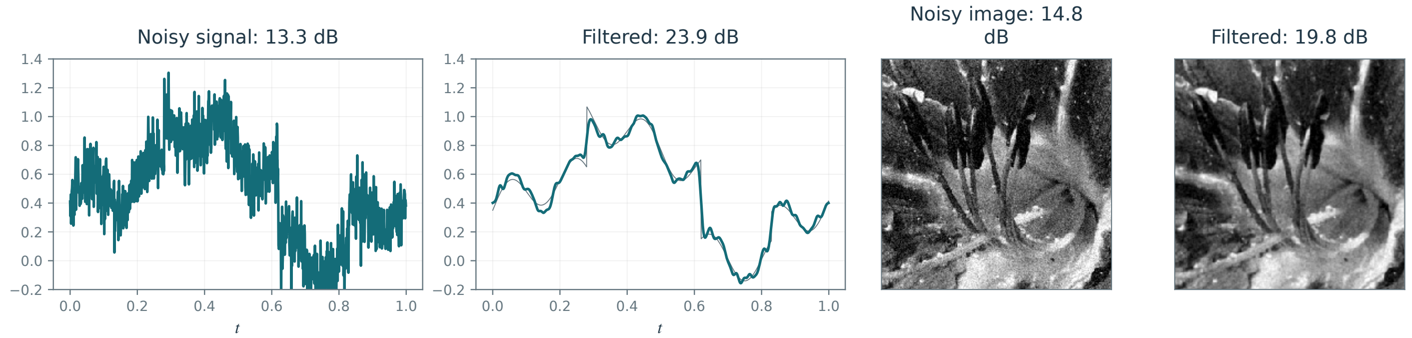
Gaussian filtering at the SNR-optimal widths. From left to right: noisy signal, filtered signal, noisy image, and filtered image.

Denoising

Current book



Proposed reconstruction



Mathematical and generation notes. Combines the previously separate signal and image comparisons on one row. Both oracle filter widths use exactly the observations and width grid of the preceding SNR plot. Clean references are used for selection and SNR measurement; the image panels share one fixed intensity range.

Code: `figures-code/chapter-denoising/filtering-optimal-1d`

Unnumbered illustration

Unnumbered illustration: cycle spin principle

Denoising

Current book

$$\begin{tikzcd} f \arrow[r, "\mathcal{D}"] \arrow[d, "\tau"'] & \mathcal{D}(f) \\ f_\tau \arrow[r, "\mathcal{D}"'] & \mathcal{D}(f_\tau) \arrow[u, "-\tau"'] \end{tikzcd}$$

Proposed reconstruction

$$\begin{tikzcd} f \arrow[r, "\mathcal{D}"] \arrow[d, "T_\tau"'] & \mathcal{D}(f) \arrow[d, "T_\tau"] \\ T_\tau f \arrow[r, "\mathcal{D}"'] & \mathcal{D}(T_\tau f) = T_\tau \mathcal{D}(f) \end{tikzcd}$$

Mathematical and generation notes. A LaTeX commutative square expresses translation equivariance: D T_tau = T_tau D, with T_tau f(x)=f(x-tau). Both paths act on the same f and use the same translation. This is the chapter property restored by cycle spinning.

Code: `figures-code/chapter-denoising/cycle-spin-principle`

Figure 8.2

Discrete operators.

Variational Priors and Regularization

Current book

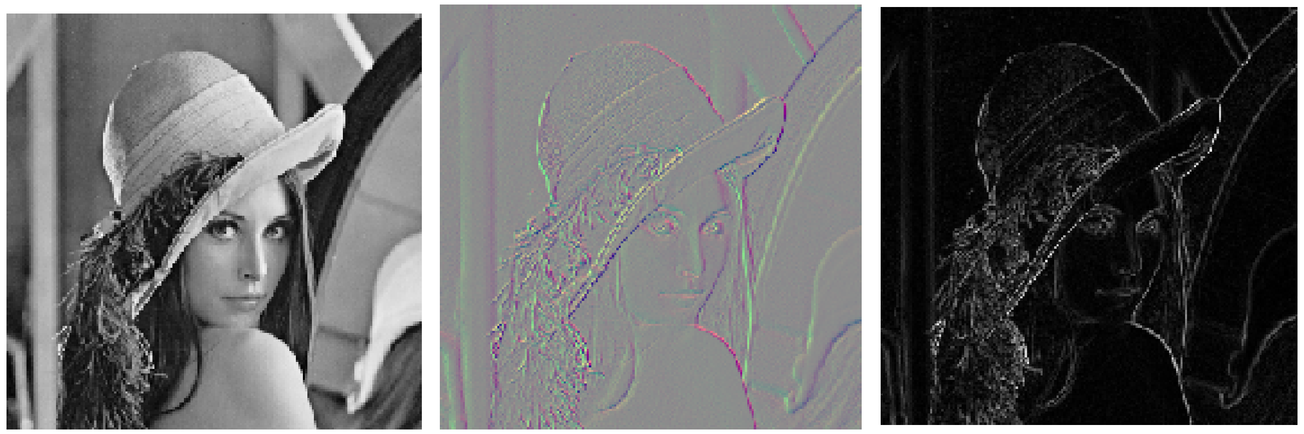
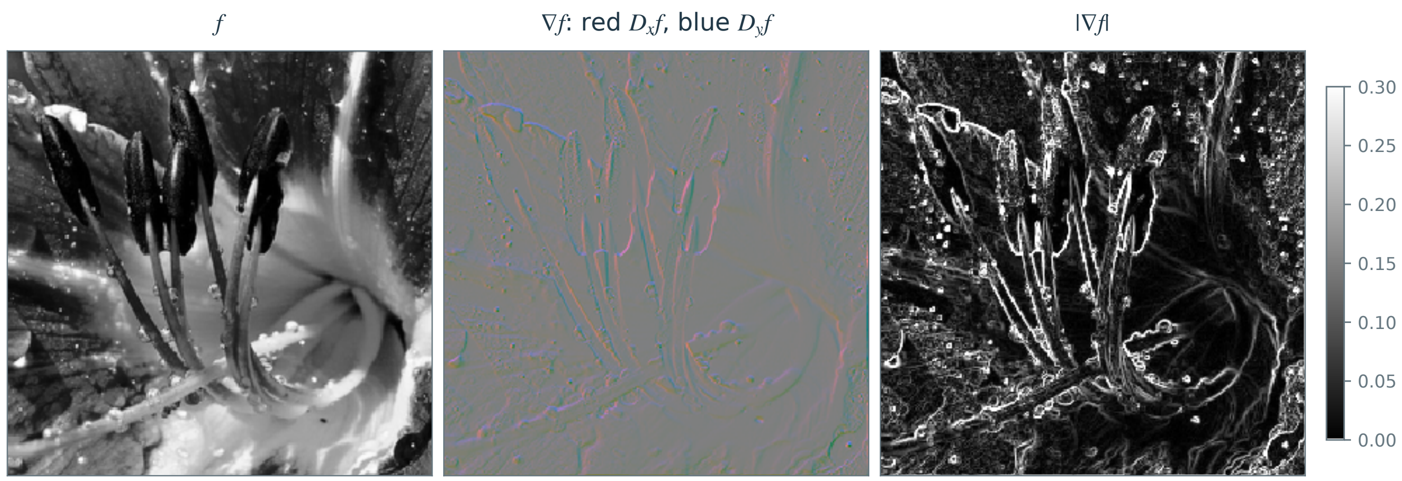


Image f | ∇f | $\|\nabla f\|$

Proposed reconstruction



Mathematical and generation notes. The middle RGB image encodes signed horizontal differences in red and signed vertical differences in blue, with a shared symmetric scale: channels are 0.5+0.5 D_xf/s and 0.5+0.5 D_yf/s, while green stays at 0.5. Zero gradient is neutral gray; increasing/decreasing channels distinguish signs. The final panel is the actual Euclidean magnitude. Periodic div=-gradient-adjoint is verified.

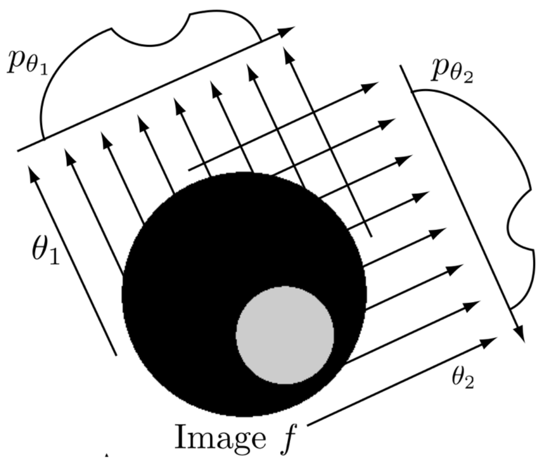
Code: `figures-code/chapter-variational-priors-and-regularization/discrete-operators`

Figure 9.7

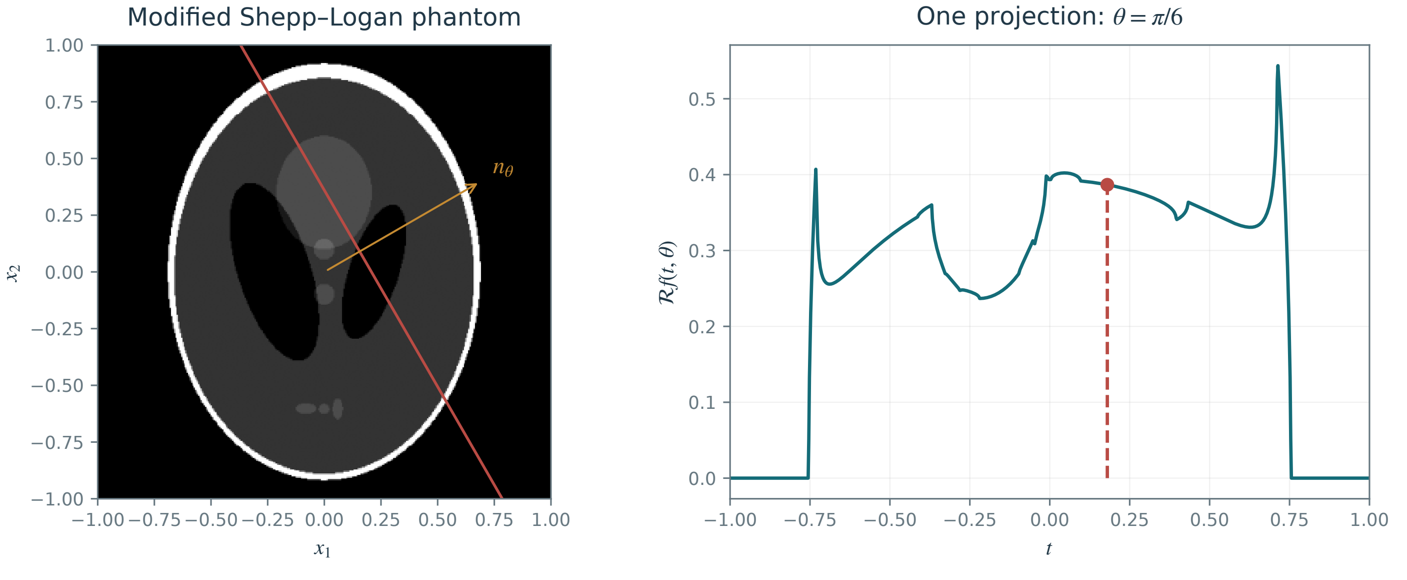
Principle of tomography acquisition.

Inverse Problems

Current book



Proposed reconstruction



Mathematical and generation notes. The conventional modified Shepp-Logan test object is generated from ten ellipses. The red line has normal n_theta and signed offset t; the red dot is its actual line integral on the exact analytic one-dimensional Radon projection. Integrating the projection recovers the phantom mass. Ellipse conventions follow the standard modified Shepp-Logan model described in https://www.mathworks.com/help/images/ref/phantom.html .

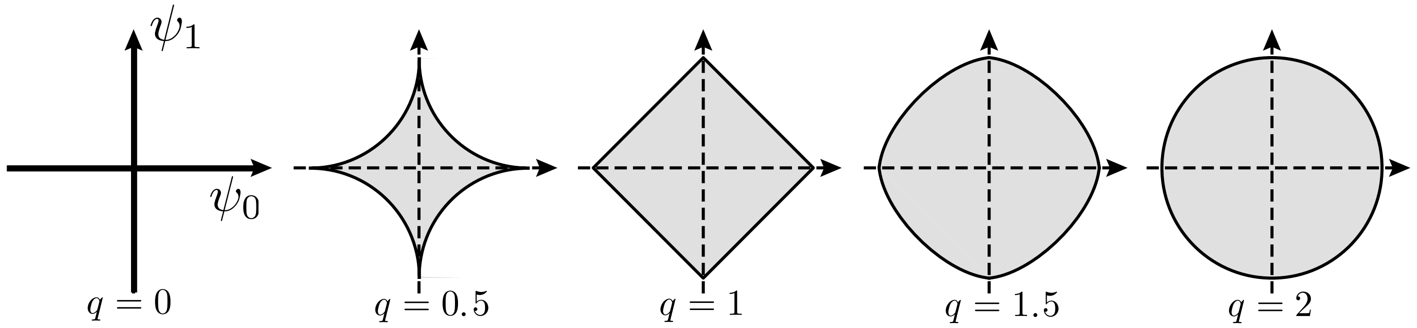
Code: `figures-code/chapter-inverse-problems/tomo-principle`

Figure 10.2

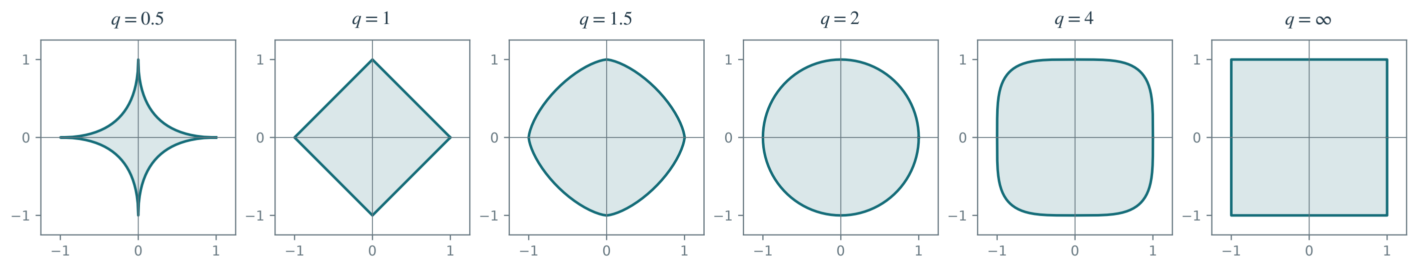
ℓ^q balls $\{x \; ; \; J_q(x) \leqslant 1\}$ for varying q.

Sparse Regularization

Current book



Proposed reconstruction



Mathematical and generation notes. Exact radial parameterizations of the two-dimensional ell-q unit sets preserve equal axis scales. The q=.5 set is nonconvex; q=1 is a diamond, q=1.5 and q=4 have intermediate strictly convex boundaries, q=2 is a disk, and q=infinity is a square.

Code: `figures-code/chapter-sparse-regularization/sparsity-lq`

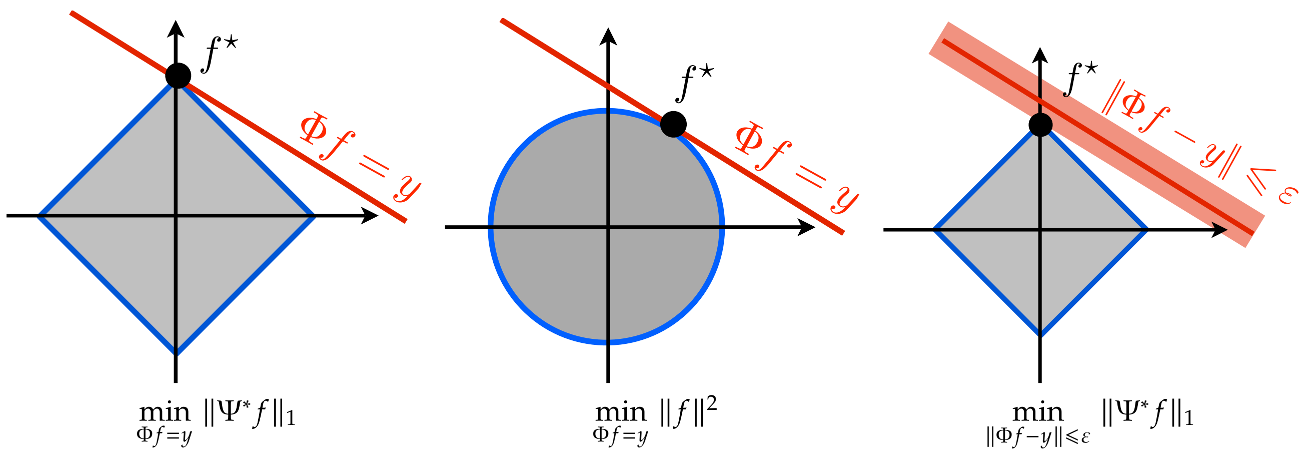
Figure 10.4

Main book, page 145

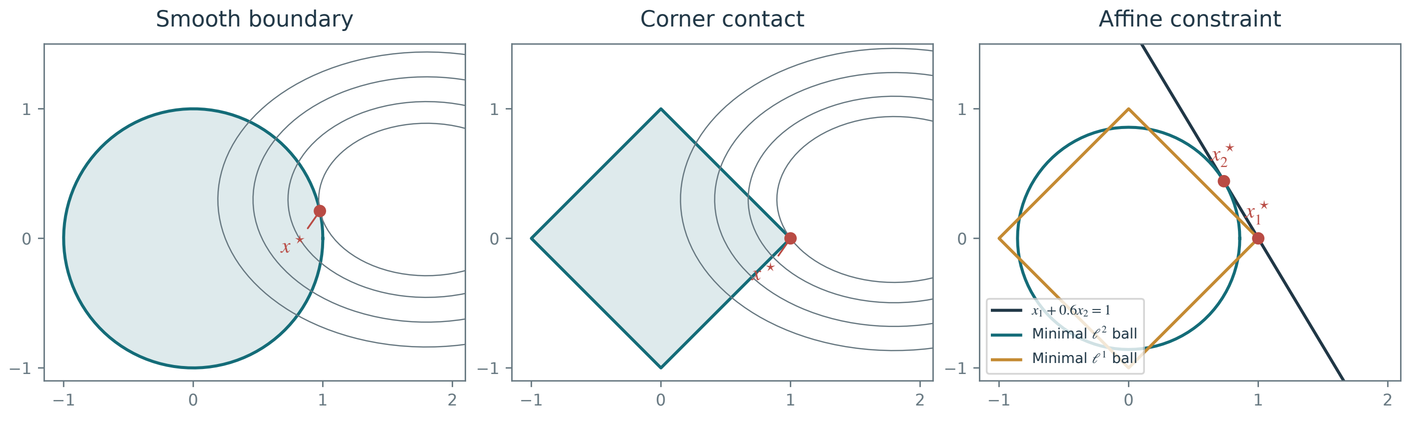
Geometry of convex optimization problems.

Sparse Regularization

Current book



Proposed reconstruction



Mathematical and generation notes. The first two panels show exact constraint contact points for F(x)=(x1-1.8)^2+2(x2-.3)^2. At the l1 corner, the supporting outward normal lies in the diamond normal cone. For x1+0.6x2=1, the minimal Euclidean ball touches at (1,.6)/1.36 and the minimal l1 ball at (1,0); both contacts and their correctly scaled balls are displayed.

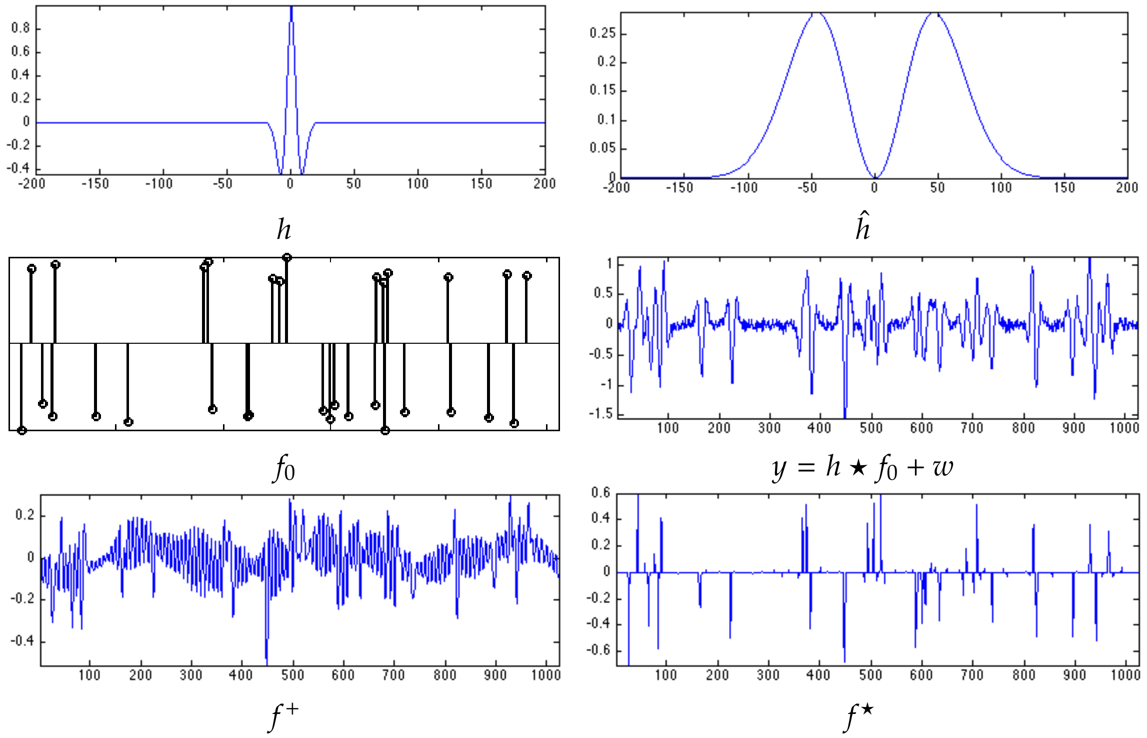
Code: `figures-code/chapter-sparse-regularization/optim-geom`

Figure 10.5

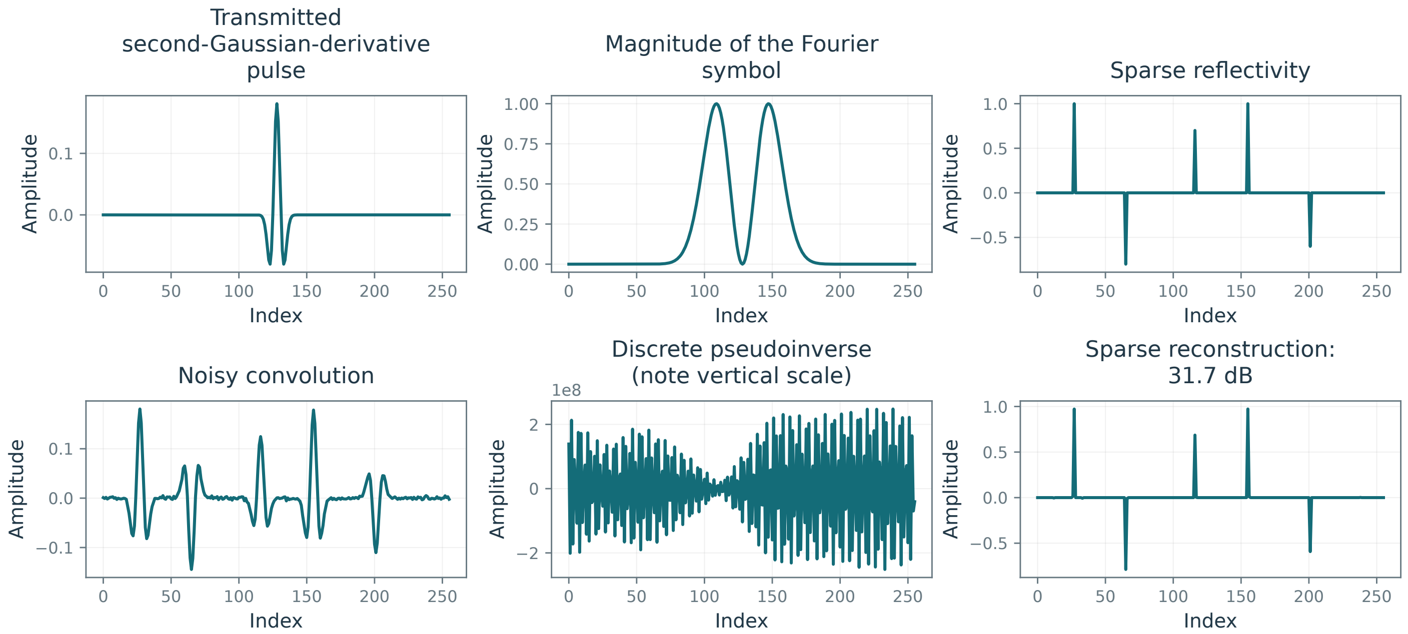
Sparse spike recovery by the pseudoinverse and by ℓ^1 regularization.

Sparse Regularization

Current book



Proposed reconstruction



Mathematical and generation notes. All six panels arise from one discrete periodic spike experiment. The pseudoinverse discards only numerically zero Fourier modes (tolerance 1e-12), so its noise amplification is visible. Sparse recovery uses the true adjoint and an oracle-selected lambda on a stated grid.

Code: `figures-code/chapter-sparse-regularization/spikes-lun`

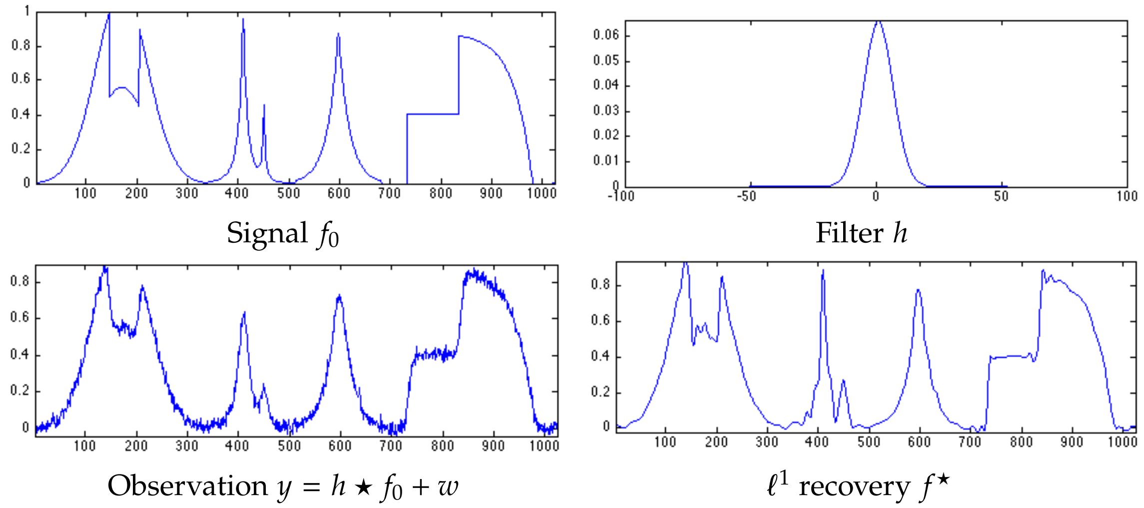
Figure 10.7

Main book, page 148

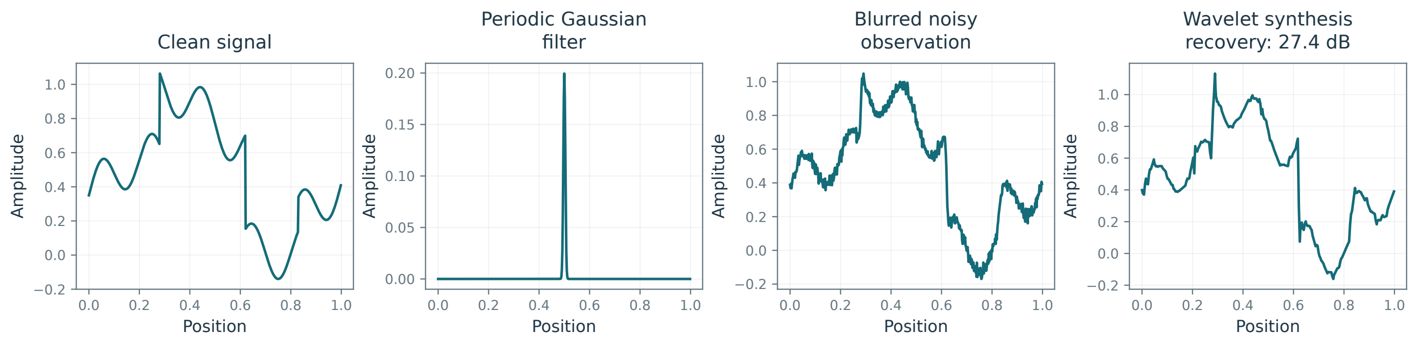
One-dimensional deconvolution with sparsity in an orthogonal wavelet basis.

Sparse Regularization

Current book



Proposed reconstruction



Mathematical and generation notes. One-dimensional deconvolution solves .5| |H Psi c-y| |^2+.005| |c| |1 in a periodic orthonormal db2 basis. The synthesis coefficients, not image values, are soft-thresholded by the optimizer. The same filter produces the observation and the forward/adjoint operators.

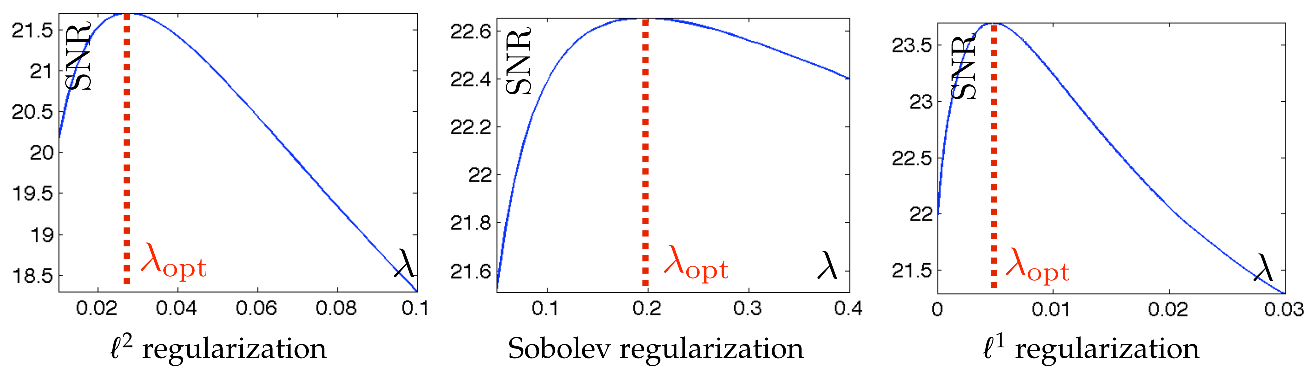
Code: `figures-code/chapter-sparse-regularization/deconvolution-1d`

Figure 10.9

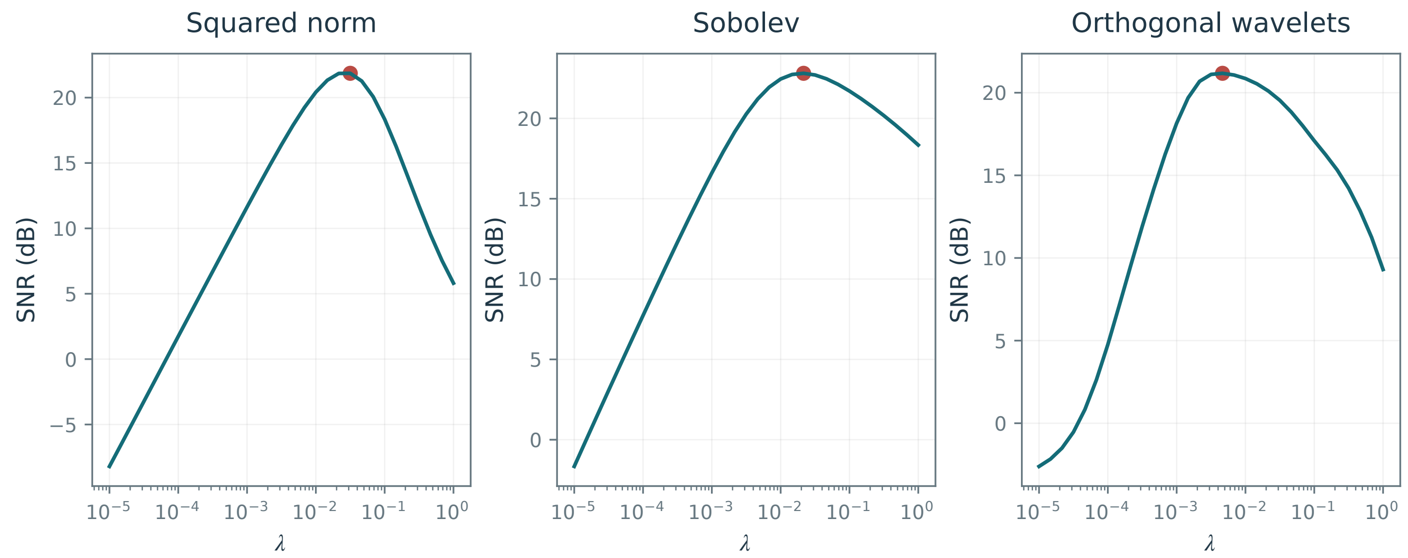
SNR as a function of λ.

Sparse Regularization

Current book



Proposed reconstruction



Mathematical and generation notes. Measured deconvolution SNR curves use the same observation, parameters, and solved objectives as Figure 10.8. Red markers are oracle maxima on the finite lambda grid, not an automatic parameter-selection method available without a clean reference.

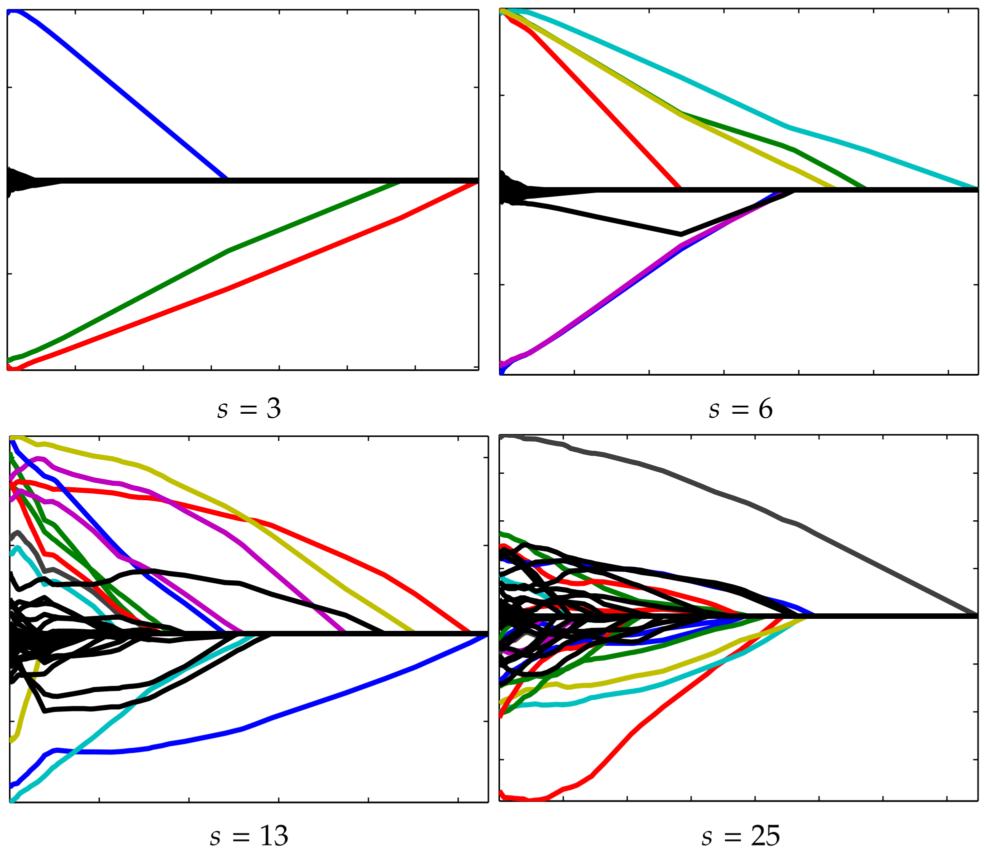
Code: `figures-code/chapter-sparse-regularization/deconvolution-2d-snr`

Figure 11.1

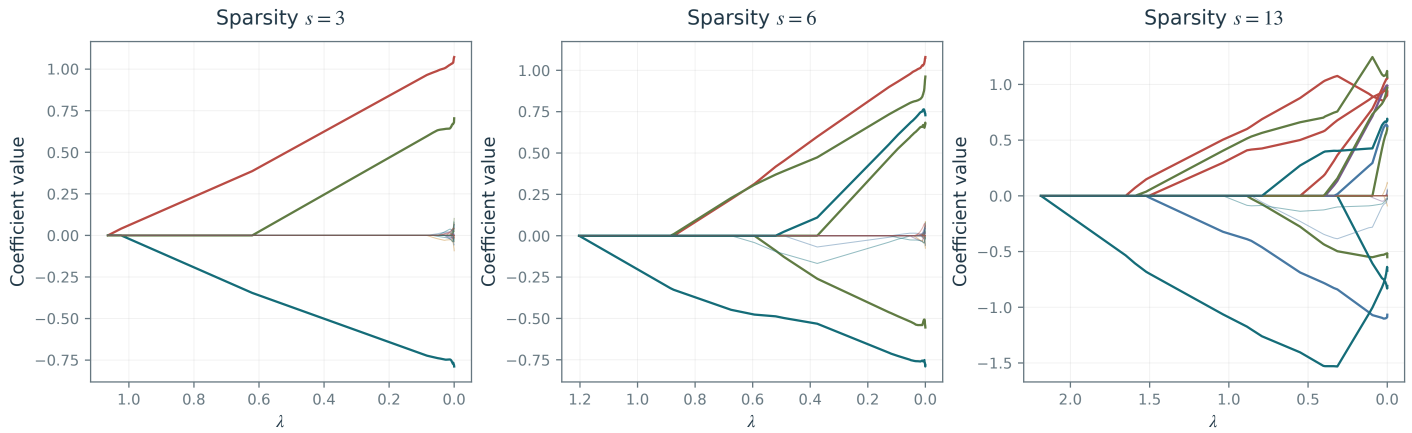
Solution paths $\lambda \mapsto x_\lambda$ for $(\mathcal{P}_\lambda(y))$.

Theory of Sparse Regularization

Current book



Proposed reconstruction



Mathematical and generation notes. Three LASSO homotopy paths use sparsities s=3,6,13, with one column-normalized sensing matrix, nested supports, and one noise realization. The library normalization by P is undone so the horizontal lambda matches .5||Ax-y||^2+lambda||x||1. Active and inactive KKT conditions are checked at every path knot.

Code: `figures-code/chapter-theory-of-sparse-regularization/homotopy`

Figure 12.1

Main book, page 167

Left: diagram of the single-pixel acquisition method. Center: ideal image $f_0 \in \mathbb{R}^Q$ in the micromirror focal plane. Right: reconstruction $f_0^\star = \Psi x^\star$ from measurements $y \in \mathbb{R}^P$, with compression factor $Q/P = 6$, using ℓ^1 regularization.

Compressed Sensing

Current book

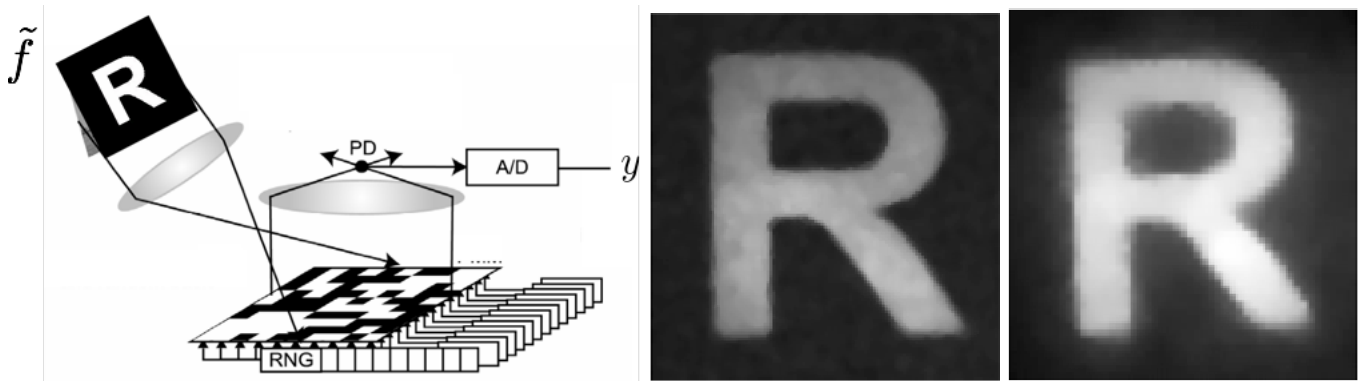
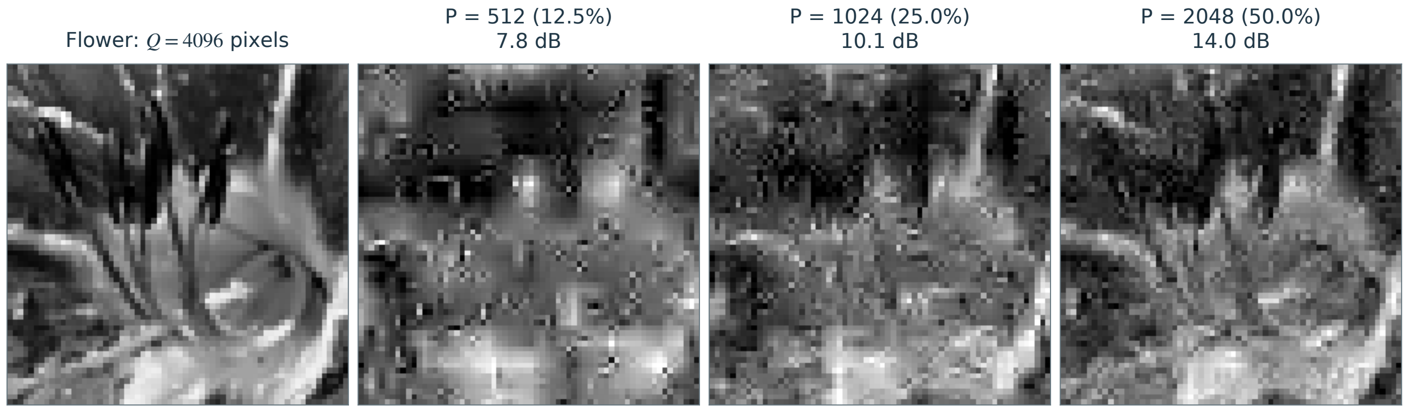


Diagram of the device

f_0

$f_0^\star, Q/P = 6$

Proposed reconstruction



Mathematical and generation notes. Simulates compressed sensing of the supplied flower with three nested measurement counts. The sensing rows are orthonormal Walsh-Hadamard patterns with a fixed random pixel permutation; signed measurements can be formed by complementary exposures. Every reconstruction solves the stated wavelet-synthesis l1 problem from its own measurements using the same lambda and iterations. The operator and adjoint are exact, with norm one. These are simulated acquisitions, not new camera measurements.

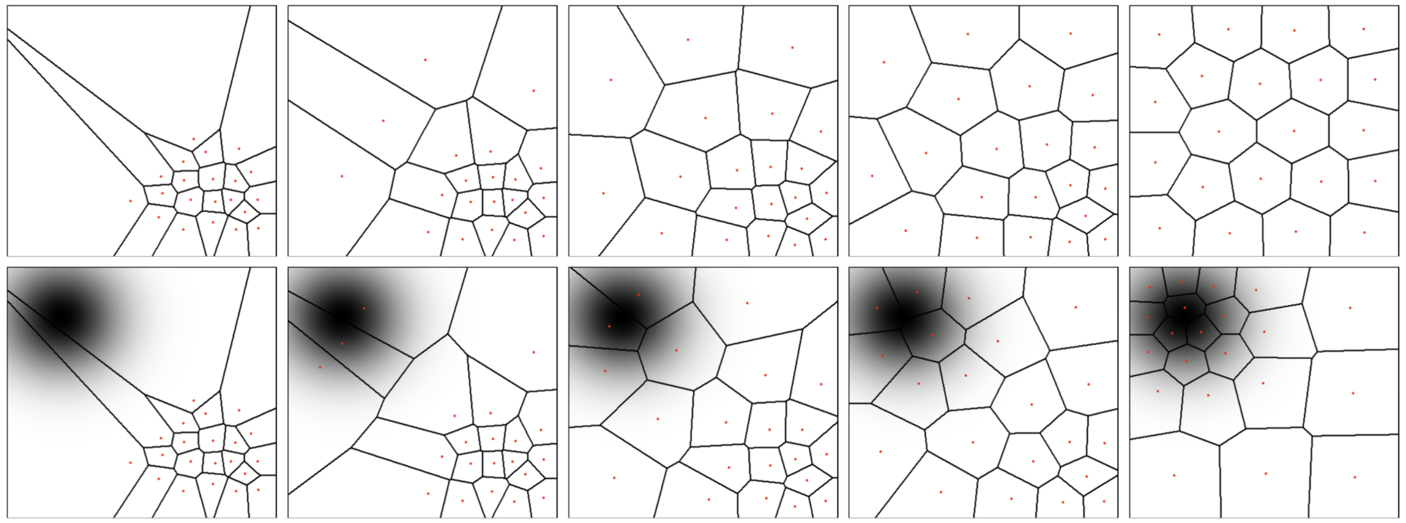
Code: `figures-code/chapter-compressed-sensing/single-pixel`

Figure 13.9

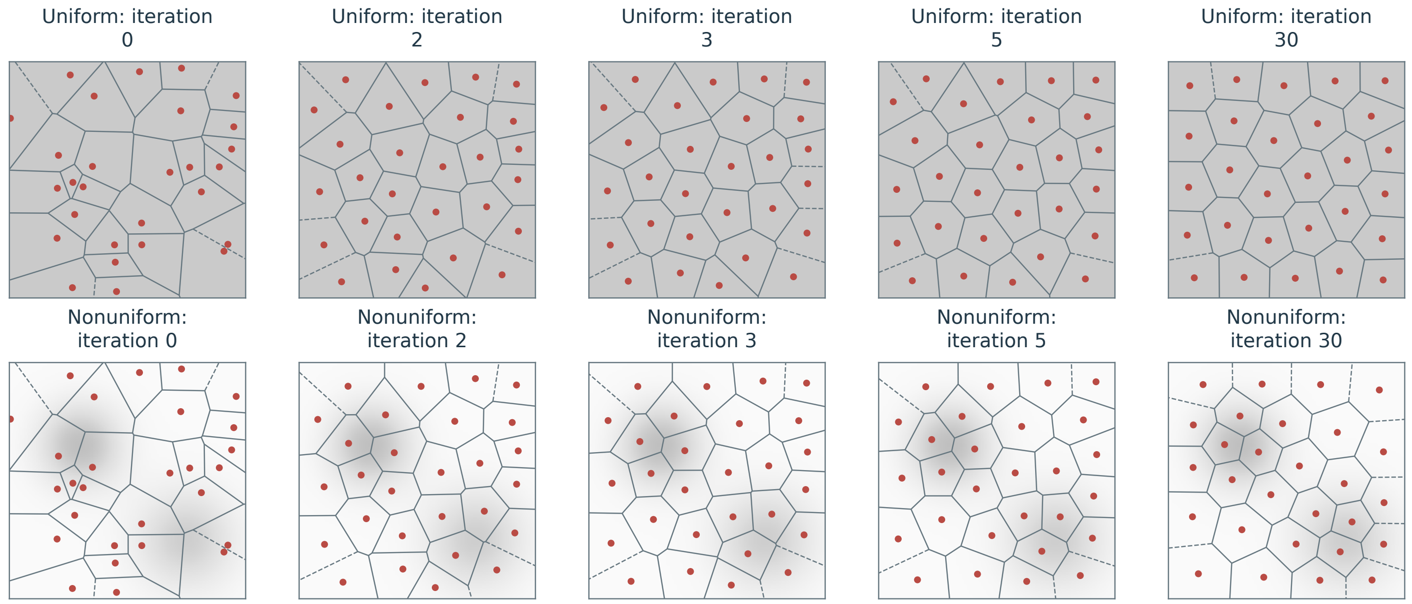
Continuous k-means (Lloyd) iterations for densities μ. Top: uniform density. Bottom: nonuniform density, shown in grayscale as the density of μ with respect to Lebesgue measure.

Basics of Machine Learning

Current book



Proposed reconstruction



Mathematical and generation notes. Continuous Lloyd updates are approximated by weighted quadrature on a 90 by 90 grid. Both rows start from the same 28 centers. Nonuniform centroids use density weights, not the unweighted mean of the Voronoi pixels. Cell boundaries are the exact Euclidean Voronoi edges of the current centroids.

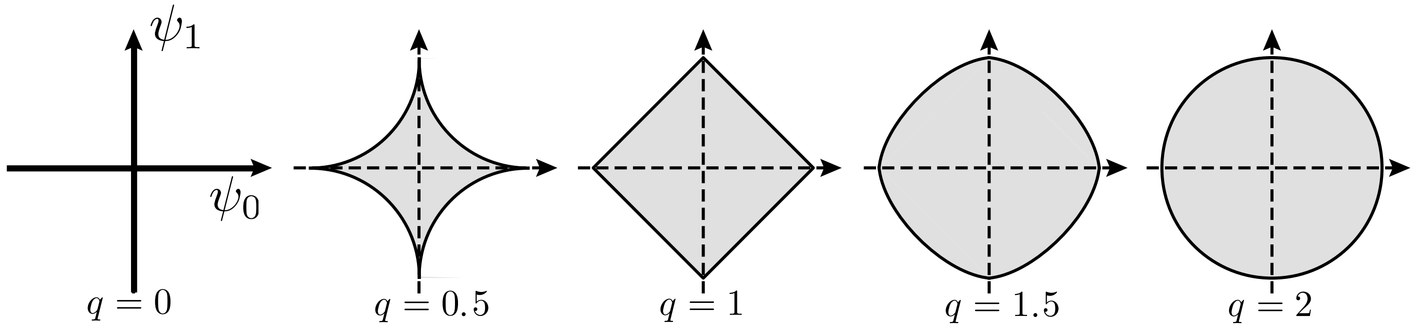
Code: `figures-code/chapter-basics-of-machine-learning/lloyd`

Figure 15.1

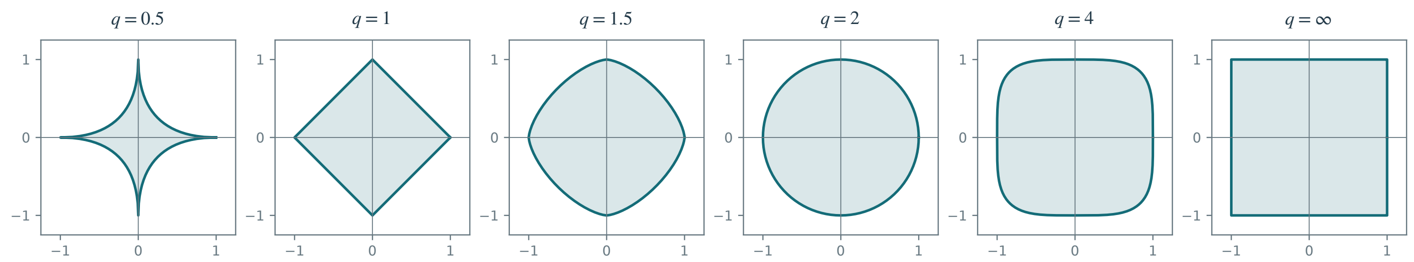
ℓ^q balls $\{x\ ;\ \sum_k |x_k|^q \leqslant 1\}$ for varying q.

Optimization & Machine Learning: Advanced Topics

Current book



Proposed reconstruction



Mathematical and generation notes. Exact radial parameterizations of the two-dimensional ell-q unit sets preserve equal axis scales. The q=.5 set is nonconvex; q=1 is a diamond, q=1.5 and q=4 have intermediate strictly convex boundaries, q=2 is a disk, and q=infinity is a square.

Code: `figures-code/chapter-optimization-and-machine-learning-advanced-topics/ml-sparsity-lq`